

Sliper QML C++ - Proje Kaynak Kodlari

Olusturulma tarihi: 2026-08-20

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CMakeLists.txt

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```
cmake_minimum_required(VERSION 3.16)
project(sliper VERSION 1.0)
```

```
set(CMAKE_CXX_STANDARD 17)
set(CMAKE_PREFIX_PATH "C:/Qt/6.7.3/mingw_64")
```

```
find_package(Qt6 REQUIRED COMPONENTS Core Quick Charts Widgets Sql PrintSupport Network)
```

```
qt_standard_project_setup()
qt_policy(SET QTP0001 NEW)
```

```
qt_add_executable(sliper
    main.cpp
    src/Backend.h
    src/Backend.cpp
    src/SensorManager.h
    src/SensorManager.cpp
    src/Calculator.h
    src/Calculator.cpp
    src/Database.h
    src/Database.cpp
    src/ReportManager.h
    src/ReportManager.cpp
    src/WifiManager.h
    src/WifiManager.cpp
)
```

```
qt_add_qml_module(sliper
    URI sliper
    VERSION 1.0
```

```
    QML_FILES
        qml/Main.qml
        qml/pages/LoginPage.qml
        qml/pages/DashboardPage.qml
        qml/pages/OlcumPage.qml
        qml/pages/SonuclarPage.qml
        qml/pages/GecmisPage.qml
        qml/pages/KalibrasyonPage.qml
        qml/components/SuTerazisi.qml
```

```
    RESOURCES
        qml/assets/logo.png
        qml/assets/egim_sensor.png
        qml/assets/loadcell.png
        qml/assets/mesafe_sensor.png
```

```
)
```

```
target_link_libraries(sliper PRIVATE
    Qt6::Core
    Qt6::Quick
    Qt6::Charts
    Qt6::Widgets
    Qt6::Sql
    Qt6::PrintSupport
```

Qt6::Network

)

main.cpp

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```
#include <QqmlContext>
#include <QApplication>
#include <QqmlApplicationEngine>
#include <QDebug>
#include <QtGlobal>
#include "src/Backend.h"
#include "src/SensorManager.h"
#include "src/Calculator.h"
#include "src/Database.h"
#include "src/ReportManager.h"
#include "src/WifiManager.h"

int main(int argc, char *argv[])
{
    qputenv("QT_QUICK_CONTROLS_STYLE", "Basic");

    QApplication app(argc, argv);
    QApplication::setApplicationName("SLIPER");

    Backend backend;
    SensorManager sensorManager;
    Calculator calculator;
    Database database;
    ReportManager reportManager;
    WifiManager wifiManager(&sensorManager, &database);

    QVariantMap mesafeUst = database.kalibrasyonGetir("mesafe_ust");
    QVariantMap mesafeAlt = database.kalibrasyonGetir("mesafe_alt");
    if (mesafeUst.value("mevcut").toBool() && mesafeAlt.value("mevcut").toBool()) {
        calculator.esikleriAyarla(mesafeUst.value("deger1").toDouble(),
                                   mesafeAlt.value("deger1").toDouble());
    }

    QqmlApplicationEngine engine;

    engine.rootContext()->setContextProperty("backend", &backend);
    engine.rootContext()->setContextProperty("sensorManager", &sensorManager);
    engine.rootContext()->setContextProperty("calculator", &calculator);
    engine.rootContext()->setContextProperty("database", &database);
    engine.rootContext()->setContextProperty("reportManager", &reportManager);
    engine.rootContext()->setContextProperty("wifiManager", &wifiManager);

    QObject::connect(
        &engine,
        &QqmlApplicationEngine::objectCreationFailed,
        &app,
        []() {
            qWarning() << "QML nesnesi olusturulamadi!";
            QCoreApplication::exit(-1);
        },
        Qt::QueuedConnection);

    engine.loadFromModule("sliper", "Main");
```

```
    return app.exec();  
}
```

src\Backend.h

13 satir

```
#pragma once
#include <QObject>

class Backend : public QObject
{
    Q_OBJECT
    Q_PROPERTY(QString durum READ durum CONSTANT)

public:
    explicit Backend(QObject *parent = nullptr);
    QString durum() const;
    Q_INVOKABLE bool girisYap(const QString &kullaniciAdi, const QString &sifre);
};
```

src\Backend.cpp

23 satir

```
#include "Backend.h"
#include <QDebug>

Backend::Backend(QObject *parent)
    : QObject(parent)
{
}

QString Backend::durum() const
{
    return "Baglanti bekleniyor...";
}

bool Backend::girisYap(const QString &kullaniciAdi, const QString &sifre)
{
    if (kullaniciAdi == "admin" && sifre == "admin123") {
        qDebug() << "Giris Basarili:" << kullaniciAdi;
        return true;
    }

    qDebug() << "Giris Basarisiz:" << kullaniciAdi;
    return false;
}
```

src\Calculator.h

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```
#pragma once
#include <QObject>

class Calculator : public QObject
{
    Q_OBJECT
    Q_PROPERTY(QString durum READ durum NOTIFY durumChanged)
    Q_PROPERTY(int strokeSayisi READ strokeSayisi NOTIFY strokeSayisiChanged)
    Q_PROPERTY(bool duraklatildi READ duraklatildi NOTIFY duraklatildiChanged)
    Q_PROPERTY(double ustEsik READ ustEsik NOTIFY esiklerChanged)
    Q_PROPERTY(double altEsik READ altEsik NOTIFY esiklerChanged)

public:
    explicit Calculator(QObject *parent = nullptr);
    QString durum() const;
    int strokeSayisi() const;
    bool duraklatildi() const;
    double ustEsik() const;
    double altEsik() const;

    Q_INVOKABLE void konumGuncelle(double konum);
    Q_INVOKABLE void duraklat();
    Q_INVOKABLE void devamEt();
    Q_INVOKABLE void sifirla();
    Q_INVOKABLE void esikleriAyarla(double ust, double alt);

signals:
    void durumChanged();
    void strokeSayisiChanged();
    void duraklatildiChanged();
    void esiklerChanged();

private:
    QString m_durum = "BEKLENIYOR";
    int m_strokeSayisi = 0;
    bool m_duraklatildi = false;
    bool m_yukaridaGorulduMu = false;
    double m_ustEsik = 450.0;
    double m_altEsik = 20.0;
};
```

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```
#include "Calculator.h"
#include <QDebug>

Calculator::Calculator(QObject *parent)
    : QObject(parent)
{
}

QString Calculator::durum() const
{
    return m_durum;
}

int Calculator::strokeSayisi() const
{
    return m_strokeSayisi;
}

bool Calculator::duraklatildi() const
{
    return m_duraklatildi;
}

double Calculator::ustEsik() const
{
    return m_ustEsik;
}

double Calculator::altEsik() const
{
    return m_altEsik;
}

void Calculator::konumGuncelle(double konum)
{
    if (m_duraklatildi) {
        return;
    }

    QString yeniDurum;

    if (konum > m_ustEsik) {
        yeniDurum = "YUKARIDA";
    } else if (konum > m_altEsik) {
        yeniDurum = "INIYOR";
    } else {
        yeniDurum = "TAMAMLANDI";
    }

    if (yeniDurum == "YUKARIDA") {
        m_yukaridaGorulduMu = true;
    }

    if (yeniDurum == "TAMAMLANDI" && m_durum != "TAMAMLANDI" && m_yukaridaGorulduMu) {
        m_strokeSayisi++;
        m_yukaridaGorulduMu = false;
    }
}
```

```

        emit strokeSayisiChanged();
    }

    if (yeniDurum != m_durum) {
        m_durum = yeniDurum;
        emit durumChanged();
    }
}

void Calculator::duraklat()
{
    if (m_duraklatildi) {
        return;
    }

    m_duraklatildi = true;
    emit duraklatildiChanged();
}

void Calculator::devamEt()
{
    if (!m_duraklatildi) {
        return;
    }

    m_duraklatildi = false;
    emit duraklatildiChanged();
}

void Calculator::sifirla()
{
    m_strokeSayisi = 0;
    m_durum = "BEKLENIYOR";
    m_duraklatildi = false;
    m_yukaridaGorulduMu = false;
    emit strokeSayisiChanged();
    emit durumChanged();
    emit duraklatildiChanged();
}

void Calculator::esikleriAyarla(double ust, double alt)
{
    if (ust <= alt) {
        qWarning() << "Gecersiz esik degerleri, ust alt'tan buyuk olmalı:" << ust << alt;
        return;
    }

    m_ustEsik = ust;
    m_altEsik = alt;
    emit esiklerChanged();
}

```

src\Database.h

31 satir

```
#pragma once
#include <QObject>
#include <QSqlDatabase>
#include <QVariantList>
#include <QVariantMap>

class Database : public QObject
{
    Q_OBJECT

public:
    explicit Database(QObject *parent = nullptr);

    Q_INVOKABLE int olcumBaslat(const QString &musteri, const QString &recete, double agirlik);
    Q_INVOKABLE void strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli);
    Q_INVOKABLE QVariantList tumOlcumleriGetir();
    Q_INVOKABLE QVariantList strokeVerileriGetir(int olcumId);
    Q_INVOKABLE QVariantMap binghamHesapla(int olcumId);
    Q_INVOKABLE bool kalibrasyonKaydet(const QString &sensor, double deger1, double deger2);
    Q_INVOKABLE QVariantMap kalibrasyonGetir(const QString &sensor);
    Q_INVOKABLE QVariantMap olcumBilgisiGetir(int olcumId);
    Q_INVOKABLE bool csvDisaAktar(int olcumId);
    Q_INVOKABLE bool loadCellNoktalarKaydet(const QVariantList &noktalar);
    Q_INVOKABLE QVariantList loadCellNoktalarGetir();
    Q_INVOKABLE bool mesafeNoktalarKaydet(const QVariantList &noktalar);
    Q_INVOKABLE QVariantList mesafeNoktalarGetir();

private:
    void tablolariOlustur();
    QSqlDatabase m_db;
};
```

```

#include "Database.h"
#include <QString>
#include <QStringError>
#include <QDebug>
#include <QDateTime>
#include <QVector>
#include <QFile>
#include <QTextStream>
#include <QStandardPaths>
#include <QDir>

Database::Database(QObject *parent)
    : QObject(parent)
{
    const QString klasor = QStandardPaths::writableLocation(QStandardPaths::AppLocalDataLocation) +
"/database";
    QDir().mkpath(klasor);

    m_db = QSqlDatabase::addDatabase("QSQLITE");
    m_db.setDatabaseName(klasor + "/sliper.db");

    if (!m_db.open()) {
        qWarning() << "Veritabani acilamadi:" << m_db.lastError().text();
    } else {
        qDebug() << "Veritabani basariyla acildi.";
        tablolariOlustur();
    }
}

void Database::tablolariOlustur()
{
    QSqlQuery sorgu;

    sorgu.exec(
        "CREATE TABLE IF NOT EXISTS Olcumler ("
        "id INTEGER PRIMARY KEY AUTOINCREMENT, "
        "tarih TEXT, "
        "musteri TEXT, "
        "recete TEXT, "
        "agirlik REAL"
        ")");

    sorgu.exec(
        "CREATE TABLE IF NOT EXISTS Strokelar ("
        "id INTEGER PRIMARY KEY AUTOINCREMENT, "
        "olcumId INTEGER, "
        "basinc REAL, "
        "konum REAL, "
        "debi REAL, "
        "gecerli INTEGER, "
        "FOREIGN KEY(olcumId) REFERENCES Olcumler(id)"
        ")");

    QSqlQuery kolonKontrol("PRAGMA table_info(Strokelar)");

```

```

bool debiKolonuVar = false;
while (kolonKontrol.next()) {
    if (kolonKontrol.value("name").toString() == "debi") {
        debiKolonuVar = true;
        break;
    }
}

if (!debiKolonuVar) {
    QSqlQuery migrasyon;
    migrasyon.exec("ALTER TABLE Strokelar ADD COLUMN debi REAL DEFAULT 0");
}

sorgu.exec(
    "CREATE TABLE IF NOT EXISTS Kalibrasyonlar ("
    "sensor TEXT PRIMARY KEY, "
    "deger1 REAL, "
    "deger2 REAL, "
    "tarih TEXT"
    ")"
);

sorgu.exec(
    "CREATE TABLE IF NOT EXISTS LoadCellNoktalari ("
    "id INTEGER PRIMARY KEY AUTOINCREMENT, "
    "hedefKg REAL, "
    "hamDeger REAL"
    ")"
);

sorgu.exec(
    "CREATE TABLE IF NOT EXISTS MesafeNoktalari ("
    "id INTEGER PRIMARY KEY AUTOINCREMENT, "
    "hedefMm REAL, "
    "hamDeger REAL"
    ")"
);

qDebug() << "Tablolar hazir.";
}

int Database::olcumBaslat(const QString &musteri, const QString &recete, double agirlik)
{
    QSqlQuery sorgu;
    sorgu.prepare(
        "INSERT INTO Olcumler (tarih, muster_i, recete, agirlik) "
        "VALUES (:tarih, :muster_i, :recete, :agirlik)"
    );

    sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
    sorgu.bindValue(":muster_i", muster_i);
    sorgu.bindValue(":recete", recete);
    sorgu.bindValue(":agirlik", agirlik);

    if (!sorgu.exec()) {
        qWarning() << "Olcum baslatilamadi:" << sorgu.lastError().text();
        return -1;
    }

    int yeniId = sorgu.lastInsertId().toInt();

```

```

        qDebug() << "Yeni olcum baslatildi, id:" << yeniId;
        return yeniId;
    }

void Database::strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli)
{
    if (olcumId <= 0) {
        qWarning() << "Gecersiz olcumId, stroke kaydedilemedi.";
        return;
    }

    QSqlQuery sorgu;
    sorgu.prepare(
        "INSERT INTO Strokelar (olcumId, basinc, konum, debi, gecerli) "
        "VALUES (:olcumId, :basinc, :konum, :debi, :gecerli)"
    );

    sorgu.bindValue(":olcumId", olcumId);
    sorgu.bindValue(":basinc", basinc);
    sorgu.bindValue(":konum", konum);
    sorgu.bindValue(":debi", debi);
    sorgu.bindValue(":gecerli", gecerli ? 1 : 0);

    if (!sorgu.exec()) {
        qWarning() << "Stroke kaydedilemedi:" << sorgu.lastError().text();
    } else {
        qDebug() << "Stroke kaydedildi, olcumId:" << olcumId;
    }
}

QVariantList Database::tumOlcumleriGetir()
{
    QVariantList sonuclar;

    QSqlQuery sorgu(
        "SELECT o.id, o.tarih, o.musteri, o.recete, o.agirlik, "
        "COUNT(s.id) AS strokeSayisi "
        "FROM Olcumler o "
        "LEFT JOIN Strokelar s ON s.olcumId = o.id "
        "GROUP BY o.id "
        "ORDER BY o.id DESC"
    );

    while (sorgu.next()) {
        QVariantMap satir;
        satir["id"] = sorgu.value("id");
        satir["tarih"] = sorgu.value("tarih");
        satir["musteri"] = sorgu.value("musteri");
        satir["recete"] = sorgu.value("recete");
        satir["agirlik"] = sorgu.value("agirlik");
        satir["strokeSayisi"] = sorgu.value("strokeSayisi");
        sonuclar.append(satir);
    }

    return sonuclar;
}

QVariantList Database::strokeVerileriGetir(int olcumId)
{
    QVariantList sonuclar;

```

```

    QSqlQuery sorgu;
    sorgu.prepare(
        "SELECT basinc, konum, debi, gecerli FROM Strokelar "
        "WHERE olcumId = :olcumId ORDER BY id ASC"
    );
    sorgu.bindValue(":olcumId", olcumId);

    if (!sorgu.exec()) {
        qWarning() << "Stroke verileri alinamadi:" << sorgu.lastError().text();
        return sonuclar;
    }

    int sira = 1;
    while (sorgu.next()) {
        QVariantMap satir;
        satir["stroke"] = sira++;
        satir["basinc"] = sorgu.value("basinc");
        satir["konum"] = sorgu.value("konum");
        satir["debi"] = sorgu.value("debi");
        satir["gecerli"] = sorgu.value("gecerli").toBool();
        sonuclar.append(satir);
    }

    return sonuclar;
}

```

```

QVariantMap Database::binghamHesapla(int olcumId)
{
    QVariantMap sonuc;

    QSqlQuery sorgu;
    sorgu.prepare(
        "SELECT basinc, debi FROM Strokelar "
        "WHERE olcumId = :olcumId AND gecerli = 1"
    );
    sorgu.bindValue(":olcumId", olcumId);

    QVector<double> pDegerleri;
    QVector<double> qDegerleri;

    if (sorgu.exec()) {
        while (sorgu.next()) {
            pDegerleri.append(sorgu.value("basinc").toDouble());
            qDegerleri.append(sorgu.value("debi").toDouble());
        }
    }

    const int n = pDegerleri.size();

    if (n < 2) {
        sonuc["tau0"] = 0.0;
        sonuc["mu"] = 0.0;
        sonuc["r2"] = 0.0;
        sonuc["yeterliVeri"] = false;
        return sonuc;
    }

    double qToplam = 0, pToplam = 0, qpToplam = 0, qKareToplam = 0;
    for (int i = 0; i < n; ++i) {

```

```

        qToplam += qDegerleri[i];
        pToplam += pDegerleri[i];
        qpToplam += qDegerleri[i] * pDegerleri[i];
        qKareToplam += qDegerleri[i] * qDegerleri[i];
    }

    const double qOrt = qToplam / n;
    const double pOrt = pToplam / n;

    const double payda = qKareToplam - n * qOrt * qOrt;
    const double mu = (payda != 0.0) ? (qpToplam - n * qOrt * pOrt) / payda : 0.0;
    const double tau0 = pOrt - mu * qOrt;

    double ssTot = 0, ssRes = 0;
    for (int i = 0; i < n; ++i) {
        const double tahmin = tau0 + mu * qDegerleri[i];
        ssRes += (pDegerleri[i] - tahmin) * (pDegerleri[i] - tahmin);
        ssTot += (pDegerleri[i] - pOrt) * (pDegerleri[i] - pOrt);
    }

    const double r2 = (ssTot != 0.0) ? (1.0 - ssRes / ssTot) : 0.0;

    sonuc["tau0"] = tau0;
    sonuc["mu"] = mu;
    sonuc["r2"] = r2;
    sonuc["yeterliVeri"] = true;
    return sonuc;
}

bool Database::kalibrasyonKaydet(const QString &sensor, double deger1, double deger2)
{
    QSqlQuery sorgu;
    sorgu.prepare(
        "INSERT INTO Kalibrasyonlar (sensor, deger1, deger2, tarih) "
        "VALUES (:sensor, :deger1, :deger2, :tarih) "
        "ON CONFLICT(sensor) DO UPDATE SET "
        "deger1 = :deger1, deger2 = :deger2, tarih = :tarih"
    );

    sorgu.bindValue(":sensor", sensor);
    sorgu.bindValue(":deger1", deger1);
    sorgu.bindValue(":deger2", deger2);
    sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));

    if (!sorgu.exec()) {
        qWarning() << "Kalibrasyon kaydedilemedi:" << sorgu.lastError().text();
        return false;
    }

    qDebug() << "Kalibrasyon kaydedildi:" << sensor;
    return true;
}

QVariantMap Database::kalibrasyonGetir(const QString &sensor)
{
    QVariantMap sonuc;

    QSqlQuery sorgu;
    sorgu.prepare("SELECT deger1, deger2, tarih FROM Kalibrasyonlar WHERE sensor = :sensor");
    sorgu.bindValue(":sensor", sensor);

```

```

        if (sorgu.exec() && sorgu.next()) {
            sonuc["deger1"] = sorgu.value("deger1");
            sonuc["deger2"] = sorgu.value("deger2");
            sonuc["tarih"] = sorgu.value("tarih");
            sonuc["mevcut"] = true;
        } else {
            sonuc["mevcut"] = false;
        }

        return sonuc;
    }

QVariantMap Database::olcumBilgisiGetir(int olcumId)
{
    QVariantMap sonuc;

    QSqlQuery sorgu;
    sorgu.prepare("SELECT tarih, musteri, recete, agirlik FROM Olcumler WHERE id = :id");
    sorgu.bindValue(":id", olcumId);

    if (sorgu.exec() && sorgu.next()) {
        sonuc["tarih"] = sorgu.value("tarih");
        sonuc["musteri"] = sorgu.value("musteri");
        sonuc["recete"] = sorgu.value("recete");
        sonuc["agirlik"] = sorgu.value("agirlik");
        sonuc["bulundu"] = true;
    } else {
        sonuc["bulundu"] = false;
    }

    return sonuc;
}

bool Database::csvDisaAktor(int olcumId)
{
    QVariantMap bilgi = olcumBilgisiGetir(olcumId);
    if (!bilgi["bulundu"].toBool()) {
        qWarning() << "Olcum bulunamadi, csv olusturulamadi.";
        return false;
    }

    QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) +
"/SliperRaporlari";
    QDir().mkpath(klasor);

    QString dosyaAdi = klasor + QString("/SLIPER_Veri_%1_%2.csv")
        .arg(olcumId)
        .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmmss"));

    QFile dosya(dosyaAdi);
    if (!dosya.open(QIODevice::WriteOnly | QIODevice::Text)) {
        qWarning() << "CSV dosyasi acilamadi:" << dosyaAdi;
        return false;
    }

    QTextStream akis(&dosya);
    akis.setEncoding(QStringConverter::Utf8);

    akis << "Olcum ID;" << olcumId << "\n";

```

```

akis << "Tarih;" << bilgi["tarih"].toString() << "\n";
akis << "Musteri;" << bilgi["musteri"].toString() << "\n";
akis << "Recete;" << bilgi["recete"].toString() << "\n";
akis << "Agirlik (kg);" << bilgi["agirlik"].toDouble() << "\n";
akis << "\n";
akis << "Stroke;Basinc (mbar);Konum (mm);Debi (m3/h);Gecerli\n";

QVariantList strokelar = strokeVerileriGetir(olcumId);
for (const QVariant &v : strokelar) {
    QVariantMap s = v.toMap();
    akis << s["stroke"].toInt() << ";";
    << s["basinc"].toDouble() << ";";
    << s["konum"].toDouble() << ";";
    << s["debi"].toDouble() << ";";
    << (s["gecerli"].toBool() ? "Evet" : "Hayir") << "\n";
}

dosya.close();
qDebug() << "CSV disa aktarildi:" << dosyaAdi;
return true;
}

bool Database::loadCellNoktalarKaydet(const QVariantList &noktalar)
{
    if (!m_db.transaction()) {
        qWarning() << "Load cell noktalarini icin transaction baslatilamadi:" << m_db.lastError().text();
        return false;
    }

    QSqlQuery temizle;
    if (!temizle.exec("DELETE FROM LoadCellNoktalar")) {
        qWarning() << "Load cell noktalarini temizlenemedi:" << temizle.lastError().text();
        m_db.rollback();
        return false;
    }

    QSqlQuery sorgu;
    sorgu.prepare(
        "INSERT INTO LoadCellNoktalar (hedefKg, hamDeger) "
        "VALUES (:hedefKg, :hamDeger)"
    );

    for (const QVariant &v : noktalar) {
        QVariantMap nokta = v.toMap();
        sorgu.bindValue(":hedefKg", nokta["hedefKg"].toDouble());
        sorgu.bindValue(":hamDeger", nokta["hamDeger"].toDouble());
        if (!sorgu.exec()) {
            qWarning() << "Load cell noktasi kaydedilemedi:" << sorgu.lastError().text();
            m_db.rollback();
            return false;
        }
    }

    if (!m_db.commit()) {
        qWarning() << "Load cell noktalarini commit edilemedi:" << m_db.lastError().text();
        return false;
    }

    qDebug() << "Load cell" << noktalar.size() << "nokta kaydedildi.";
    return true;
}

```

```

}

QVariantList Database::loadCellNoktalariGetir()
{
    QVariantList sonuclar;

    QSqlQuery sorgu("SELECT hedefKg, hamDeger FROM LoadCellNoktalari ORDER BY hamDeger ASC");

    while (sorgu.next()) {
        QVariantMap satir;
        satir["hedefKg"] = sorgu.value("hedefKg");
        satir["hamDeger"] = sorgu.value("hamDeger");
        sonuclar.append(satir);
    }

    return sonuclar;
}

bool Database::mesafeNoktalariKaydet(const QVariantList &noktalar)
{
    if (!m_db.transaction()) {
        qWarning() << "Mesafe noktaları için transaction başlatılamadı:" << m_db.lastError().text();
        return false;
    }

    QSqlQuery temizle;
    if (!temizle.exec("DELETE FROM MesafeNoktalari")) {
        qWarning() << "Mesafe noktaları temizlenemedi:" << temizle.lastError().text();
        m_db.rollback();
        return false;
    }

    QSqlQuery sorgu;
    sorgu.prepare(
        "INSERT INTO MesafeNoktalari (hedefMm, hamDeger) "
        "VALUES (:hedefMm, :hamDeger)"
    );

    for (const QVariant &v : noktalar) {
        QVariantMap nokta = v.toMap();
        sorgu.bindValue(":hedefMm", nokta["hedefMm"].toDouble());
        sorgu.bindValue(":hamDeger", nokta["hamDeger"].toDouble());
        if (!sorgu.exec()) {
            qWarning() << "Mesafe noktasi kaydedilemedi:" << sorgu.lastError().text();
            m_db.rollback();
            return false;
        }
    }

    if (!m_db.commit()) {
        qWarning() << "Mesafe noktaları commit edilemedi:" << m_db.lastError().text();
        return false;
    }

    qDebug() << "Mesafe" << noktalar.size() << "nokta kaydedildi.";
    return true;
}

QVariantList Database::mesafeNoktalariGetir()
{

```

```
QVariantList sonuclar;

QStringQuery sorgu("SELECT hedefMm, hamDeger FROM MesafeNoktalari ORDER BY hamDeger ASC");

while (sorgu.next()) {
    QVariantMap satir;
    satir["hedefMm"] = sorgu.value("hedefMm");
    satir["hamDeger"] = sorgu.value("hamDeger");
    sonuclar.append(satir);
}

return sonuclar;
}
```

```
src\ReportManager.h
```

12 satir

[illegible]

```

#include "ReportManager.h"
#include <QPrinter>
#include <QPainter>
#include <QTextDocument>
#include <QStandardPaths>
#include <QDir>
#include <QDateTime>
#include <QDebug>

ReportManager::ReportManager(QObject *parent)
    : QObject(parent)
{
}

bool ReportManager::pdfOlustur(int olcumId, const QString &musteri, const QString &recete,
                                double tau0, double mu, double r2)
{
    QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) +
"/SliperRaporlari";
    QDir().mkpath(klasor);

    QString dosyaAdi = klasor + QString("/SLIPER_Rapor_%1_%2.pdf")
        .arg(olcumId)
        .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HH:mm:ss"));

    QTextDocument belge;
    QString html = QString(
        "<h1 style='color:#3b82f6;'>SLIPER Analiz Raporu</h1>"
        "<p><b>Olcum ID:</b> %1</p>"
        "<p><b>Musteri:</b> %2</p>"
        "<p><b>Recete:</b> %3</p>"
        "<p><b>Tarih:</b> %4</p>"
        "<hr>"
        "<h2>Bingham Model Sonuclari</h2>"
        "<p><b>Akma Gerilmesi (tau0):</b> %5 mbar</p>"
        "<p><b>Plastik Viskozite (mu):</b> %6 mbar*h/m3</p>"
        "<p><b>Uyum Kalitesi (R2):</b> %7</p>"
        "<hr>"
        "<p style='color:#6b7280; font-size:10px;'>Liya Laboratuvar Test Cihazlari - SLIPER Analiz
Sistemi</p>"
    ).arg(olcumId)
        .arg(musteri)
        .arg(recete)
        .arg(QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"))
        .arg(QString::number(tau0, 'f', 2))
        .arg(QString::number(mu, 'f', 2))
        .arg(QString::number(r2, 'f', 2));

    belge.setHtml(html);

    QPrinter yazici(QPrinter::HighResolution);
    yazici.setOutputFormat(QPrinter::PdfFormat);
    yazici.setOutputFileName(dosyaAdi);

    belge.print(&yazici);

```

```
qDebug() << "PDF rapor olusturuldu:" << dosyaAdi;  
return true;  
}
```

src\SensorManager.h

57 satir

```
#pragma once
#include <QObject>

class SensorManager : public QObject
{
    Q_OBJECT
    Q_PROPERTY(double basinc READ basinc NOTIFY basincChanged)
    Q_PROPERTY(double konum READ konum NOTIFY konumChanged)
    Q_PROPERTY(double hiz READ hiz NOTIFY hizChanged)
    Q_PROPERTY(double debi READ debi NOTIFY debiChanged)
    Q_PROPERTY(double egimX READ egimX NOTIFY egimXChanged)
    Q_PROPERTY(double egimY READ egimY NOTIFY egimYChanged)
    Q_PROPERTY(double hamAgirlik READ hamAgirlik NOTIFY hamAgirlikChanged)
    Q_PROPERTY(double hamMesafe READ hamMesafe NOTIFY hamMesafeChanged)
    Q_PROPERTY(bool veriGecerli READ veriGecerli NOTIFY veriGecerliChanged)

public:
    explicit SensorManager(QObject *parent = nullptr);

    double basinc() const;
    double konum() const;
    double hiz() const;
    double debi() const;
    double egimX() const;
    double egimY() const;
    double hamAgirlik() const;
    double hamMesafe() const;
    bool veriGecerli() const;

    Q_INVOKABLE void veriGuncelle(double basinc, double konum, double hiz, double debi, double egimX, double
egimY);
    Q_INVOKABLE void hamAgirlikGuncelle(double hamAgirlik);
    Q_INVOKABLE void hamMesafeGuncelle(double hamMesafe);
    Q_INVOKABLE void veriyiGecersizYap();

signals:
    void basincChanged();
    void konumChanged();
    void hizChanged();
    void debiChanged();
    void egimXChanged();
    void egimYChanged();
    void hamAgirlikChanged();
    void hamMesafeChanged();
    void veriGecerliChanged();
    void veriGuncellendi();

private:
    double m_basinc = 0.0;
    double m_konum = 0.0;
    double m_hiz = 0.0;
    double m_debi = 0.0;
    double m_egimX = 0.0;
    double m_egimY = 0.0;
    double m_hamAgirlik = 0.0;
    double m_hamMesafe = 0.0;
```

```
    bool m_veriGecerli = false;  
};
```

src\SensorManager.cpp

55 satir

```
#include "SensorManager.h"
```

```
SensorManager::SensorManager(QObject *parent)
    : QObject(parent)
{
}
```

```
double SensorManager::basinc() const { return m_basinc; }
double SensorManager::konum() const { return m_konum; }
double SensorManager::hiz() const { return m_hiz; }
double SensorManager::debi() const { return m_debi; }
double SensorManager::egimX() const { return m_egimX; }
double SensorManager::egimY() const { return m_egimY; }
double SensorManager::hamAgirlik() const { return m_hamAgirlik; }
double SensorManager::hamMesafe() const { return m_hamMesafe; }
bool SensorManager::veriGecerli() const { return m_veriGecerli; }
```

```
void SensorManager::veriGuncelle(double basinc, double konum, double hiz, double debi, double egimX, double
egimY)
```

```
{
    const bool veriGecerliDegisti = !m_veriGecerli;
    m_veriGecerli = true;

    if (m_basinc != basinc) { m_basinc = basinc; emit basincChanged(); }
    if (m_konum != konum) { m_konum = konum; emit konumChanged(); }
    if (m_hiz != hiz) { m_hiz = hiz; emit hizChanged(); }
    if (m_debi != debi) { m_debi = debi; emit debiChanged(); }
    if (m_egimX != egimX) { m_egimX = egimX; emit egimXChanged(); }
    if (m_egimY != egimY) { m_egimY = egimY; emit egimYChanged(); }
    if (veriGecerliDegisti) { emit veriGecerliChanged(); }

    emit veriGuncellendi();
}
```

```
void SensorManager::hamAgirlikGuncelle(double hamAgirlik)
```

```
{
    if (m_hamAgirlik != hamAgirlik) {
        m_hamAgirlik = hamAgirlik;
        emit hamAgirlikChanged();
    }
}
```

```
void SensorManager::hamMesafeGuncelle(double hamMesafe)
```

```
{
    if (m_hamMesafe != hamMesafe) {
        m_hamMesafe = hamMesafe;
        emit hamMesafeChanged();
    }
}
```

```
void SensorManager::veriyiGecersizYap()
```

```
{
    if (!m_veriGecerli) return;
    m_veriGecerli = false;
}
```

```
    emit veriGecerliChanged();  
}
```

src\WifiManager.h

75 satir

```
#pragma once
#include <QObject>
#include <QTcpSocket>
#include <QElapsedTimer>
#include <QVector>
#include "SensorManager.h"
#include "Database.h"

class WifiManager : public QObject
{
    Q_OBJECT
    Q_PROPERTY(bool baglandi READ baglandi NOTIFY baglandiChanged)
    Q_PROPERTY(bool baglaniyor READ baglaniyor NOTIFY baglaniyorChanged)
    Q_PROPERTY(QString durumMesaji READ durumMesaji NOTIFY durumMesajiChanged)

public:
    explicit WifiManager(SensorManager *sensorManager, Database *database, QObject *parent = nullptr);

    bool baglandi() const;
    bool baglaniyor() const;
    QString durumMesaji() const;

    Q_INVOKABLE void baglan();
    Q_INVOKABLE void baglantiyiKes();
    Q_INVOKABLE void kalibrasyonYenidenYukle();

signals:
    void baglandiChanged();
    void baglaniyorChanged();
    void durumMesajiChanged();

private slots:
    void soketBaglandi();
    void soketAyrildi();
    void soketHata(QAbstractSocket::SocketError hata);
    void veriHazir();

private:
    void durumGuncelle(const QString &mesaj);
    void jsonSatiriIsle(const QByteArray &satir);
    void egimHesapla(double accelX, double accelY, double accelZ, double &egimX, double &egimY) const;
    void kalibrasyonYukle();
    double loadCellInterpolasyon(double hamDeger) const;
    double mesafeInterpolasyon(double hamDeger) const;

    QTcpSocket *m_soket = nullptr;
    SensorManager *m_sensorManager = nullptr;
    Database *m_database = nullptr;
    QByteArray m_tamponVeri;

    bool m_baglandi = false;
    bool m_baglaniyor = false;
    QString m_durumMesaji = "Bağlı Değil";

    bool m_ilkPaket = true;
    double m_oncekiKonum = 0.0;
```

```
qint64 m_oncekiZamanMs = 0;
QElapsedTimer m_zamanlayici;

// --- Kayitli kalibrasyon degerleri (Database'den yuklenir) ---

QVector<double> m_loadCellHamDegerleri;
QVector<double> m_loadCellKiloDegerleri;

QVector<double> m_mesafeHamDegerleri;
QVector<double> m_mesafeMmDegerleri;

double m_egimOffsetX = 0.0;
double m_egimOffsetY = 0.0;

static constexpr const char *ESP32_IP = "192.168.4.1";
static constexpr quint16 ESP32_PORT = 8888;

static constexpr double BORU_CAPI_M = 0.126;
};
```

src\WifiManager.cpp

268 satir

```
#include "WifiManager.h"
#include <QJsonDocument>
#include <QJsonObject>
#include <QDebug>
#include <cmath>

WifiManager::WifiManager(SensorManager *sensorManager, Database *database, QObject *parent)
    : QObject(parent)
    , m_sensorManager(sensorManager)
    , m_database(database)
{
    m_soket = new QTcpSocket(this);

    connect(m_soket, &QTcpSocket::connected, this, &WifiManager::soketBaglandi);
    connect(m_soket, &QTcpSocket::disconnected, this, &WifiManager::soketAyrildi);
    connect(m_soket, &QTcpSocket::errorOccurred, this, &WifiManager::soketHata);
    connect(m_soket, &QTcpSocket::readyRead, this, &WifiManager::veriHazir);
}

bool WifiManager::baglandi() const { return m_baglandi; }
bool WifiManager::baglaniyor() const { return m_baglaniyor; }
QString WifiManager::durumMesaji() const { return m_durumMesaji; }

void WifiManager::baglan()
{
    if (m_baglandi || m_baglaniyor) return;

    m_baglaniyor = true;
    emit baglaniyorChanged();
    durumGuncelle("SLIPER-ESP32'ye baglaniliyor...");

    m_soket->connectToHost(QString(ESP32_IP), ESP32_PORT);
}

void WifiManager::baglantiyiKes()
{
    m_soket->disconnectFromHost();
}

void WifiManager::kalibrasyonYenidenYukle()
{
    kalibrasyonYukle();
}

void WifiManager::kalibrasyonYukle()
{
    if (!m_database) return;

    m_loadCellHamDegerleri.clear();
    m_loadCellKiloDegerleri.clear();
    const QVariantList noktalar = m_database->loadCellNoktalariniGetir();
    for (const QVariant &v : noktalar) {
        QVariantMap n = v.toMap();
        m_loadCellHamDegerleri.append(n["hamDeger"].toDouble());
        m_loadCellKiloDegerleri.append(n["hedefKg"].toDouble());
    }
}
```

```

qDebug() << "Load cell" << m_loadCellHamDegerleri.size() << "nokta yuklendi.";

m_mesafeHamDegerleri.clear();
m_mesafeMmDegerleri.clear();
const QVariantList mesafeNoktaları = m_database->mesafeNoktalarıGetir();
for (const QVariant &v : mesafeNoktaları) {
    QVariantMap n = v.toMap();
    m_mesafeHamDegerleri.append(n["hamDeger"].toDouble());
    m_mesafeMmDegerleri.append(n["hedefMm"].toDouble());
}
qDebug() << "Mesafe" << m_mesafeHamDegerleri.size() << "nokta yuklendi.";

QVariantMap egim = m_database->kalibrasyonGetir("egim");
if (egim.value("mevcut").toBool()) {
    m_egimOffsetX = egim.value("deger1").toDouble();
    m_egimOffsetY = egim.value("deger2").toDouble();
    qDebug() << "Egim kalibrasyonu yuklendi: X=" << m_egimOffsetX << "Y=" << m_egimOffsetY;
}
}

void WifiManager::soketBaglandi()
{
    m_baglandi = true;
    m_baglaniyor = false;
    emit baglandiChanged();
    emit baglaniyorChanged();
    durumGuncelle("SLIPER-ESP32 Bağlı");

    m_ilkPaket = true;
    m_zamanlayici.restart();
    kalibrasyonYukle();

    if (m_sensorManager) {
        m_sensorManager->veriyiGecersizYap();
    }

    qDebug() << "ESP32'ye baglanildi:" << ESP32_IP << ESP32_PORT;
}

void WifiManager::soketAyrildi()
{
    m_baglandi = false;
    m_baglaniyor = false;
    emit baglandiChanged();
    emit baglaniyorChanged();
    durumGuncelle("Bağlı Değil");

    if (m_sensorManager) {
        m_sensorManager->veriyiGecersizYap();
    }

    qDebug() << "ESP32 baglantisi kesildi.";
}

void WifiManager::soketHata(QAbstractSocket::SocketError hata)
{
    m_baglandi = false;
    m_baglaniyor = false;
    emit baglandiChanged();
    emit baglaniyorChanged();
}

```

```

        durumGuncelle("Baglanti hatasi: " + m_soket->errorString());

        qWarning() << "Wifi soket hatasi:" << hata << m_soket->errorString();
    }

void WifiManager::veriHazar()
{
    m_tamponVeri.append(m_soket->readAll());

    int satirSonu;
    while ((satirSonu = m_tamponVeri.indexOf('\n')) != -1) {
        QByteArray satir = m_tamponVeri.left(satirSonu).trimmed();
        m_tamponVeri.remove(0, satirSonu + 1);

        if (!satir.isEmpty()) {
            jsonSatiriIsle(satir);
        }
    }
}

void WifiManager::egimHesapla(double accelX, double accelY, double accelZ, double &egimX, double &egimY)
const
{
    egimX = std::atan2(accelY, accelZ) * 180.0 / M_PI;
    egimY = std::atan2(-accelX, std::sqrt(accelY * accelY + accelZ * accelZ)) * 180.0 / M_PI;
}

void WifiManager::jsonSatiriIsle(const QByteArray &satir)
{
    QJsonParseError hata;
    QJsonDocument belge = QJsonDocument::fromJson(satir, &hata);

    if (hata.error != QJsonParseError::NoError || !belge.isObject()) {
        qWarning() << "JSON parse hatasi:" << hata.errorString() << satir;
        return;
    }

    QJsonObject obj = belge.object();

    const double hamAgirlik = obj.value("hamAgirlik").toDouble();
    const double hamMesafe = obj.contains("hamMesafe")
        ? obj.value("hamMesafe").toDouble()
        : obj.value("konum").toDouble();
    const double accelX = obj.value("accelX").toDouble();
    const double accelY = obj.value("accelY").toDouble();
    const double accelZ = obj.value("accelZ").toDouble();

    // Kalibrasyon ekraninda gerekiyor: ham deger her zaman guncellensin
    if (m_sensorManager) {
        m_sensorManager->hamAgirlikGuncelle(hamAgirlik);
        m_sensorManager->hamMesafeGuncelle(hamMesafe);
    }

    const double konum = m_mesafeHamDegerleri.isEmpty()
        ? hamMesafe
        : mesafeInterpolasyon(hamMesafe);

    // --- Basinc: kayitli kalibrasyon ile ---
    const double agirlikKg = loadCellInterpolasyon(hamAgirlik);
    const double basinc = agirlikKg * 98.1;

```

```

// --- Hiz ve debi: konum farkindan turetiliyor ---
double hiz = 0.0;
double debi = 0.0;

const qint64 simdikiZamanMs = m_zamanlayici.elapsed();

if (!m_ilkPaket) {
    double dt = (simdikiZamanMs - m_oncekiZamanMs) / 1000.0;
    if (dt <= 0.0) dt = 0.001;

    const double boruAlaniM2 = M_PI * (BORU_CAPI_M / 2.0) * (BORU_CAPI_M / 2.0);
    hiz = std::fabs(konum - m_oncekiKonum) / 1000.0 / dt;
    debi = hiz * boruAlaniM2 * 3600.0;
} else {
    m_ilkPaket = false;
}

m_oncekiKonum = konum;
m_oncekiZamanMs = simdikiZamanMs;

// --- Egim: ham ivmeden aci, sonra kayitli ofset uygulanir ---
double egimXHam = 0.0;
double egimYHam = 0.0;
egimHesapla(accelX, accelY, accelZ, egimXHam, egimYHam);

const double egimX = egimXHam - m_egimOfsetX;
const double egimY = egimYHam - m_egimOfsetY;

if (m_sensorManager) {
    m_sensorManager->veriGuncelle(basinc, konum, hiz, debi, egimX, egimY);
}
}

void WifiManager::durumGuncelle(const QString &mesaj)
{
    if (m_durumMesaji != mesaj) {
        m_durumMesaji = mesaj;
        emit durumMesajiChanged();
    }
}

double WifiManager::loadCellInterpolasyon(double hamDeger) const
{
    const int n = m_loadCellHamDegerleri.size();
    if (n == 0) return 0.0;
    if (n == 1) return m_loadCellKiloDegerleri[0];

    int i;
    if (hamDeger <= m_loadCellHamDegerleri[0]) {
        i = 0;
    } else if (hamDeger >= m_loadCellHamDegerleri[n - 1]) {
        i = n - 2;
    } else {
        for (i = 0; i < n - 1; ++i) {
            if (hamDeger <= m_loadCellHamDegerleri[i + 1]) break;
        }
    }

    const double ham1 = m_loadCellHamDegerleri[i];

```

```

const double ham2 = m_loadCellHamDegerleri[i + 1];
const double kg1 = m_loadCellKiloDegerleri[i];
const double kg2 = m_loadCellKiloDegerleri[i + 1];

if (ham2 == ham1) return kg1;
return kg1 + (hamDeger - ham1) * (kg2 - kg1) / (ham2 - ham1);
}

```

```

double WifiManager::mesafeInterpolasyon(double hamDeger) const
{
    const int n = m_mesafeHamDegerleri.size();
    if (n == 0) return 0.0;
    if (n == 1) return m_mesafeMmDegerleri[0];

    int i;
    if (hamDeger <= m_mesafeHamDegerleri[0]) {
        i = 0;
    } else if (hamDeger >= m_mesafeHamDegerleri[n - 1]) {
        i = n - 2;
    } else {
        for (i = 0; i < n - 1; ++i) {
            if (hamDeger <= m_mesafeHamDegerleri[i + 1]) break;
        }
    }

    const double ham1 = m_mesafeHamDegerleri[i];
    const double ham2 = m_mesafeHamDegerleri[i + 1];
    const double mm1 = m_mesafeMmDegerleri[i];
    const double mm2 = m_mesafeMmDegerleri[i + 1];

    if (ham2 == ham1) return mm1;
    return mm1 + (hamDeger - ham1) * (mm2 - mm1) / (ham2 - ham1);
}

```

qml\Main.qml

28 satir

```
import QtQuick 6.7
import QtQuick.Controls 6.7
import QtQuick.Window 6.7
import QtQuick.Layouts 6.7
import "pages"

ApplicationWindow {
    id: pencere
    visibility: Window.Maximized
    title: "SLIPER Analiz"

    background: Rectangle {
        color: "#060d17"
    }

    StackLayout {
        id: anaStack
        anchors.fill: parent
        currentIndex: 0

        LoginPage {
            onGirisBasarili: anaStack.currentIndex = 1
        }

        DashboardPage {
        }
    }
}
```

78 satir

```
import QtQuick 6.7

Item {
    id: root
    property real egimX: 0
    property real egimY: 0
    property real maxAci: 15

    width: 200
    height: 200

    readonly property real yaricap: (width / 2) - 20
    readonly property bool duz: Math.abs(egimX) < 0.5 && Math.abs(egimY) < 0.5

    readonly property real kabarcikOffsetX: Math.max(-1, Math.min(1, egimY / maxAci)) * yaricap
    readonly property real kabarcikOffsetY: Math.max(-1, Math.min(1, egimX / maxAci)) * yaricap

    Rectangle {
        anchors.fill: parent
        radius: width / 2
        color: "#0f1420"
        border.color: "#1e2a3f"
        border.width: 2
    }

    Rectangle {
        anchors.centerIn: parent
        width: root.yaricap * 1.2
        height: width
        radius: width / 2
        color: "transparent"
        border.color: "#1e2a3f"
        border.width: 1
    }

    Rectangle {
        anchors.centerIn: parent
        width: parent.width - 20
        height: 1
        color: "#1e2a3f"
    }

    Rectangle {
        anchors.centerIn: parent
        width: 1
        height: parent.height - 20
        color: "#1e2a3f"
    }

    Rectangle {
        id: kabarcik
        width: 28
        height: 28
        radius: 14
        color: root.duz ? "#16a34a" : "#dc2626"
        border.color: root.duz ? "#4ade80" : "#f87171"
```

```
border.width: 2
```

```
x: (root.width / 2) - (width / 2) + root.kabarcikOffsetX  
y: (root.height / 2) - (height / 2) + root.kabarcikOffsetY
```

```
Behavior on x { NumberAnimation { duration: 150; easing.type: Easing.OutQuad } }  
Behavior on y { NumberAnimation { duration: 150; easing.type: Easing.OutQuad } }  
Behavior on color { ColorAnimation { duration: 200 } }  
Behavior on border.color { ColorAnimation { duration: 200 } }
```

```
}
```

```
Text {  
    anchors.top: parent.bottom  
    anchors.horizontalCenter: parent.horizontalCenter  
    anchors.topMargin: 8  
    text: root.duz ? "Düz" : "Ayarlanıyor..."  
    color: root.duz ? "#4ade80" : "#6b7280"  
    font.family: "Segoe UI"  
    font.pixelSize: 12  
    font.bold: root.duz
```

```
}
```

```
}
```

199 satır

```

import QtQuick 6.7
import QtQuick.Controls 6.7
import QtQuick.Layouts 6.7

Rectangle {
    color: "#0a0e17"

    readonly property var navOgeleri: [
        { etiket: "Ölçüm", ikon: "📊" },
        { etiket: "Sonuçlar", ikon: "📈" },
        { etiket: "Geçmiş", ikon: "🕒" },
        { etiket: "Kalibrasyon", ikon: "⚙️" }
    ]

    Row {
        anchors.fill: parent
        spacing: 0

        Rectangle {
            id: sidebar
            width: 230
            height: parent.height

            gradient: Gradient {
                GradientStop { position: 0.0; color: "#121b30" }
                GradientStop { position: 1.0; color: "#080b12" }
            }

            Image {
                anchors.top: parent.top
                anchors.horizontalCenter: parent.horizontalCenter
                anchors.topMargin: 24
                source: "../assets/logo.png"
                width: 144
                height: 144
                fillMode: Image.PreserveAspectFit
            }

            Rectangle {
                id: baglantiDurumu
                anchors.top: parent.top
                anchors.left: parent.left
                anchors.right: parent.right
                anchors.leftMargin: 24
                anchors.rightMargin: 24
                anchors.topMargin: 178
                height: 36
                radius: 6
                color: baglantiAlani.hoverli ? "#101a2c" : "transparent"
                Behavior on color { ColorAnimation { duration: 120 } }

                Row {
                    anchors.left: parent.left
                    anchors.leftMargin: 8
                    anchors.verticalCenter: parent.verticalCenter
                    spacing: 8

```

```

        Rectangle {
            width: 8
            height: 8
            radius: 4
            anchors.verticalCenter: parent.verticalCenter
            color: wifiManager.baglandi ? "#16a34a" : (wifiManager.baglaniyor ? "#f59e0b" :
"#dc2626")
        }

        Text {
            text: wifiManager.baglandi ? "SLIPER-ESP32 Bağlı" : (wifiManager.baglaniyor ?
"Bağlanıyor..." : "Bağlı Değil")
            color: "#9ca3af"
            font.family: "Segoe UI"
            font.pixelSize: 12
            anchors.verticalCenter: parent.verticalCenter
        }
    }

    MouseArea {
        id: baglantiAlani
        anchors.fill: parent
        hoverEnabled: true
        cursorShape: Qt.PointingHandCursor
        enabled: !wifiManager.baglandi && !wifiManager.baglaniyor
        onClicked: wifiManager.baglan()
    }
}

Rectangle {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.leftMargin: 24
    anchors.rightMargin: 24
    anchors.topMargin: 224
    height: 1
    color: "#1f2c46"
}

Column {
    anchors.top: parent.top
    anchors.topMargin: 242
    width: parent.width
    spacing: 2

    Repeater {
        model: navOgeleri

        Rectangle {
            id: navOgesi
            width: parent.width
            height: 46
            property bool aktif: icerikStack.currentIndex === index
            property bool hoverli: false
            color: aktif ? "#182644" : (hoverli ? "#101a2c" : "transparent")
            Behavior on color { ColorAnimation { duration: 120 } }

            Rectangle {

```

```

        width: 3
        height: parent.height
        color: "#3b82f6"
        visible: navOgesi.aktif
    }

    Row {
        anchors.left: parent.left
        anchors.leftMargin: 24
        anchors.verticalCenter: parent.verticalCenter
        spacing: 12

        Text {
            text: modelData.ikon
            font.pixelSize: 15
            anchors.verticalCenter: parent.verticalCenter
        }

        Text {
            text: modelData.etiket
            color: navOgesi.aktif ? "#6ea8ff" : (navOgesi.hoverli ? "#dce8f5" :
"#9ca3af")

            font.family: "Segoe UI"
            font.pixelSize: 14
            font.bold: navOgesi.aktif
            anchors.verticalCenter: parent.verticalCenter
            Behavior on color { ColorAnimation { duration: 120 } }
        }
    }

    MouseArea {
        anchors.fill: parent
        hoverEnabled: true
        cursorShape: Qt.PointingHandCursor
        onEntered: navOgesi.hoverli = true
        onExited: navOgesi.hoverli = false
        onClicked: icerikStack.currentIndex = index
    }
}

}

}

}

Rectangle {
    id: sidebarAyraci
    width: 3
    height: parent.height
    color: "#05070b"

    Rectangle {
        anchors.right: parent.right
        anchors.top: parent.top
        anchors.bottom: parent.bottom
        width: 1
        color: "#403b82f6"
    }
}

StackLayout {
    id: icerikStack

```

```
width: parent.width - sidebar.width - sidebarAyraci.width
height: parent.height
currentIndex: 0
```

```
OlcumPage {
  onOlcumTamamlandi: function(id) {
    sonuclarSayfasi.olcumId = id
    icerikStack.currentIndex = 1
  }
}
```

```
SonuclarPage {
  id: sonuclarSayfasi
}
```

```
GecmisPage {
  onTestSecildi: function(id) {
    sonuclarSayfasi.olcumId = id
    icerikStack.currentIndex = 1
  }
}
```

```
KalibrasyonPage {}
```

```
}
```

```
}
```

```
}
```

201 satir

```
import QtQuick 6.7
import QtQuick.Controls 6.7

Rectangle {
    id: gecmisSayfasi
    color: "#0a0e17"

    signal testSecildi(int id)

    property var olcumListesi: []

    function listeyiYenile() {
        olcumListesi = database.tumOlcumleriGetir()
    }

    onVisibleChanged: {
        if (visible) {
            listeyiYenile()
        }
    }

    Component.onCompleted: listeyiYenile()

    Column {
        anchors.fill: parent
        anchors.margins: 20
        spacing: 16

        Row {
            width: parent.width
            spacing: 12

            TextField {
                id: aramaKutusu
                width: parent.width - filtreKutusu.width - parent.spacing
                height: 40
                placeholderText: "Müşteri veya reçete ara..."
                placeholderTextColor: "#4b5563"
                color: "ffffff"
                font.pixelSize: 13
                leftPadding: 34
                verticalAlignment: TextInput.AlignVCenter
                background: Rectangle {
                    color: "#0f1420"
                    radius: 8
                    border.color: aramaKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
                    border.width: 1
                }

                Text {
                    anchors.left: parent.left
                    anchors.leftMargin: 12
                    anchors.verticalCenter: parent.verticalCenter
                    text: "🔍"
                    color: "#4b5563"
                    font.pixelSize: 13
                }
            }
        }
    }
}
```

```

    }
}

ComboBox {
    id: filtreKutusu
    width: 160
    height: 40
    model: ["Tüm Testler", "Son 7 Gün", "Son 30 Gün"]

    background: Rectangle {
        color: "#0f1420"
        radius: 8
        border.color: filtreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
        border.width: 1
    }

    contentItem: Text {
        text: filtreKutusu.displayText
        color: "#9ca3af"
        font.pixelSize: 13
        leftPadding: 10
        verticalAlignment: Text.AlignVCenter
    }
}

}

ListView {
    width: parent.width
    height: parent.height - 56
    clip: true
    spacing: 10

    model: gecmisSayfasi.olcumListesi

    delegate: Rectangle {
        id: gecmisKarti
        width: parent.width
        height: 76
        radius: 10
        color: "#0f1420"
        border.color: kartAlani.containsMouse ? "#3b82f6" : "#1e2a3f"
        border.width: 1
        Behavior on border.color { ColorAnimation { duration: 120 } }

        MouseArea {
            id: kartAlani
            anchors.fill: parent
            hoverEnabled: true
            acceptedButtons: Qt.NoButton
        }

        Rectangle {
            id: rozet
            width: 44
            height: 44
            radius: 10
            anchors.left: parent.left
            anchors.verticalCenter: parent.verticalCenter
            anchors.leftMargin: 16
            color: "#132335"

```

```

Text {
    anchors.centerIn: parent
    text: modelData.musteri.charAt(0).toUpperCase()
    color: "#3b82f6"
    font.family: "Segoe UI"
    font.pixelSize: 18
    font.bold: true
}
}

Column {
    anchors.left: rozet.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.leftMargin: 14
    spacing: 4

    Row {
        spacing: 8

        Text {
            text: modelData.tarih
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 11
        }

        Rectangle {
            width: strokeMetni.implicitWidth + 12
            height: 16
            radius: 8
            color: "#2a2010"
            anchors.verticalCenter: parent.verticalCenter

            Text {
                id: strokeMetni
                anchors.centerIn: parent
                text: modelData.strokeSayisi + " stroke"
                color: "#e8a020"
                font.family: "Segoe UI"
                font.pixelSize: 10
                font.bold: true
            }
        }
    }
}

Text {
    text: "Müşteri: " + modelData.musteri + " • Reçete: " + modelData.recete
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}
}

Button {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.rightMargin: 16
    height: 34

```

```
text: "Görüntüle"  
font.pixelSize: 12
```

```
onClicked: testSecildi(modelData.id)
```

```
background: Rectangle {  
    radius: 6  
    color: parent.hovered ? "#4f8cf7" : "#3b82f6"  
    Behavior on color { ColorAnimation { duration: 120 } }  
}
```

```
contentItem: Text {  
    text: parent.text  
    color: "#ffffff"  
    font: parent.font  
    horizontalAlignment: Text.AlignHCenter  
    verticalAlignment: Text.AlignVCenter  
    leftPadding: 8  
    rightPadding: 8  
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

2960 satir

```
import QtQuick 6.7
import QtQuick.Controls 6.7
import QtQuick.Layouts 6.7
import "../components"

Rectangle {
    color: "#0a0e17"

    property int seciliSensor: -1
    property int egimAdimi: 0
    property int loadCellAdimi: 0
    property int mesafeAdimi: 0

    property bool egimKayitliVar: false
    property string egimSonTarih: ""

    property bool loadCellKayitliVar: false
    property string loadCellSonTarih: ""
    property int loadCellNoktaSayisi: 0
    property var loadCellNoktalari: []
    onLoadCellNoktalariChanged: loadCellAdimi = loadCellNoktalari.length
    property int loadCellToplamNokta: 10
    property bool loadCellSayiSecildi: false
    property int loadCellDuzenlemeIndex: -1

    property bool loadCellAraEkranGoster: false
    property bool loadCellGoruntuleModu: false
    property bool loadCellTamDuzenlemeModu: false
    property bool loadCellYenidenOnayGoster: false
    property bool loadCellTamamlandiOnayGoster: false

    property bool loadCellBildirimGoster: false
    property bool loadCellBildirimBasarili: true
    property string loadCellBildirimMesaj: ""

    property bool mesafeUstKayitliVar: false
    property bool mesafeAltKayitliVar: false
    property string mesafeSonTarih: ""
    property real mesafeUstDeger: 0.0
    property real mesafeAltDeger: 0.0
    property bool mesafeBAraEkranGoster: false
    property bool mesafeYakinUyariGoster: false

    property int mesafeAsama: 0
    property bool mesafeOlcumKayitliVar: false
    property int mesafeOlcumNoktaSayisi: 0
    property var mesafeNoktalari: []
    property int mesafeToplamNokta: 10
    property bool mesafeSayiSecildi: false
    property int mesafeDuzenlemeIndex: -1

    property bool mesafeAraEkranGoster: false
    property bool mesafeGoruntuleModu: false
    property bool mesafeTamDuzenlemeModu: false
    property bool mesafeYenidenOnayGoster: false
    property bool mesafeTamamlandiOnayGoster: false
```

```

property bool mesafeBildirimGoster: false
property bool mesafeBildirimBasarili: true
property string mesafeBildirimMesaj: ""

function loadCellSiraliKopya(dizi) {
    var kopya = dizi.slice()
    kopya.sort(function(a, b) { return a.hamDeger - b.hamDeger })
    return kopya
}

function loadCellKaydetVeBildir() {
    var sirali = loadCellSiraliKopya(loadCellNoktalar)
    var basarili = database.loadCellNoktalarKaydet(sirali)
    loadCellBildirimBasarili = basarili
    loadCellBildirimMesaj = basarili ? "✓ Kalibrasyon kaydedildi" : "✗ Kalibrasyon kaydedilemedi, tekrar
deneyin"
    loadCellBildirimGoster = true
    loadCellBildirimTimer.restart()
}

Timer {
    id: loadCellBildirimTimer
    interval: loadCellBildirimBasarili ? 1800 : 2500
    repeat: false
    onTriggered: {
        loadCellBildirimGoster = false
        if (loadCellBildirimBasarili) {
            seciliSensor = -1
        }
    }
}

function mesafeSiraliKopya(dizi) {
    var kopya = dizi.slice()
    kopya.sort(function(a, b) { return a.hamDeger - b.hamDeger })
    return kopya
}

function mesafeKaydetVeBildir() {
    var sirali = mesafeSiraliKopya(mesafeNoktalar)
    var basarili = database.mesafeNoktalarKaydet(sirali)
    if (basarili) {
        wifiManager.kalibrasyonYenidenYukle()
    }
    mesafeBildirimBasarili = basarili
    mesafeBildirimMesaj = basarili ? "✓ Kalibrasyon kaydedildi" : "✗ Kalibrasyon kaydedilemedi, tekrar
deneyin"
    mesafeBildirimGoster = true
    mesafeBildirimTimer.restart()
}

function mesafeBKaydetVeBitir() {
    var basarili1 = database.kalibrasyonKaydet("mesafe_ust", mesafeUstDeger, 0.0)
    var basarili2 = database.kalibrasyonKaydet("mesafe_alt", mesafeAltDeger, 0.0)
    var basarili = basarili1 && basarili2
    if (basarili) {
        calculator.esikleriAyarla(mesafeUstDeger, mesafeAltDeger)
    }
    mesafeBildirimBasarili = basarili

```

```

mesafeBildirimMesaj = basarili ? "✓ Stroke sınırları kaydedildi" : "× Kaydedilemedi, tekrar deneyin"
mesafeBildirimGoster = true
mesafeBildirimTimer.restart()
}

Timer {
    id: mesafeBildirimTimer
    interval: mesafeBildirimBasarili ? 1800 : 2500
    repeat: false
    onTriggered: {
        mesafeBildirimGoster = false
        if (mesafeBildirimBasarili) {
            if (mesafeAsama === 0) {
                mesafeAsama = 1
            } else {
                seciliSensor = -1
            }
        }
    }
}

}

onSeciliSensorChanged: {
    if (seciliSensor === 0) {
        var e = database.kalibrasyonGetir("egim")
        egimKayitliVar = e.mevcut
        egimSonTarih = e.mevcut ? e.tarih : ""
    } else if (seciliSensor === 1) {
        var noktalar = database.loadCellNoktalarGetir()
        loadCellKayitliVar = noktalar.length > 0
        loadCellNoktaSayisi = noktalar.length
        loadCellNoktalar = []
        loadCellSayiSecildi = false
        loadCellDuzenlemeIndex = -1
        loadCellTamDuzenlemeModu = false
        loadCellGoruntuleModu = false
        loadCellYenidenOnayGoster = false
        loadCellTamamlandiOnayGoster = false
        loadCellBildirimGoster = false
        loadCellAraEkranGoster = loadCellKayitliVar
    } else if (seciliSensor === 2) {
        var mNoktalar = database.mesafeNoktalarGetir()
        mesafeOlcumKayitliVar = mNoktalar.length > 0
        mesafeOlcumNoktaSayisi = mNoktalar.length
        mesafeNoktalar = []
        mesafeSayiSecildi = false
        mesafeDuzenlemeIndex = -1
        mesafeTamDuzenlemeModu = false
        mesafeGoruntuleModu = false
        mesafeYenidenOnayGoster = false
        mesafeTamamlandiOnayGoster = false
        mesafeBildirimGoster = false
        mesafeAraEkranGoster = mesafeOlcumKayitliVar

        var u = database.kalibrasyonGetir("mesafe_ust")
        var a = database.kalibrasyonGetir("mesafe_alt")
        mesafeUstKayitliVar = u.mevcut
        mesafeAltKayitliVar = a.mevcut
        mesafeSonTarih = u.mevcut ? u.tarih : (a.mevcut ? a.tarih : "")
        mesafeUstDeger = u.mevcut ? u.deger1 : 0.0
        mesafeAltDeger = a.mevcut ? a.deger1 : 0.0
    }
}

```

```

mesafeBAraEkranGoster = mesafeUstKayitliVar && mesafeAltKayitliVar
mesafeYakinUyariGoster = false
mesafeAdimi = 0

mesafeAsama = mesafeOlcumKayitliVar ? 1 : 0
}
}

StackLayout {
    anchors.fill: parent
    currentIndex: seciliSensor === -1 ? 0 : 1

    Item {
        Text {
            anchors.top: parent.top
            anchors.left: parent.left
            anchors.margins: 20
            text: "KALİBRASYON – Sensör Seçin"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 12
            font.bold: true
            font.letterSpacing: 1
        }

        Row {
            anchors.centerIn: parent
            spacing: 24

            Rectangle {
                id: egimKarti
                width: 350
                height: 450
                radius: 12
                color: "#0f1420"
                border.color: egimKarti.hovered ? "#14b8a6" : "#1e2a3f"
                border.width: 1
                property bool hovered: false
                scale: hovered ? 1.03 : 1.0
                Behavior on border.color { ColorAnimation { duration: 120 } }
                Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }

                Column {
                    anchors.centerIn: parent
                    spacing: 14

                    Item {
                        anchors.horizontalCenter: parent.horizontalCenter
                        width: 210
                        height: 150

                        Image {
                            anchors.centerIn: parent
                            width: parent.width
                            height: parent.height
                            source: "../assets/egim_sensor.png"
                            fillMode: Image.PreserveAspectFit
                        }
                    }
                }
            }
        }
    }
}

```

```
Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Eğim Sensörü";
color: "#dce8f5"; font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
```

```
Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Yatay hizalama
kalibrasyonu"; color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }
```

```
Rectangle {
    anchors.horizontalCenter: parent.horizontalCenter
    visible: egimKayitliVar
    width: kaliMetni1.implicitWidth + 16
    height: 20
    radius: 10
    color: "#123321"
```

```
Text {
    id: kaliMetni1
    anchors.centerIn: parent
    text: "✓ Kalibre Edildi"
    color: "#4ade80"
    font.pixelSize: 10
    font.bold: true
}
```

```
}
```

```
MouseArea {
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onEntered: egimKarti.hovered = true
    onExited: egimKarti.hovered = false
    onClicked: { seciliSensor = 0; egimAdimi = 0 }
}
```

```
}
```

```
Rectangle {
    id: loadCellKarti
    width: 350
    height: 450
    radius: 12
    color: "#0f1420"
    border.color: loadCellKarti.hovered ? "#3b82f6" : "#1e2a3f"
    border.width: 1
    property bool hovered: false
    scale: hovered ? 1.03 : 1.0
    Behavior on border.color { ColorAnimation { duration: 120 } }
    Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
```

```
Column {
    anchors.centerIn: parent
    spacing: 14
```

```
Item {
    anchors.horizontalCenter: parent.horizontalCenter
    width: 210
    height: 150
```

```
Image {
    anchors.centerIn: parent
    width: parent.width
    height: parent.height
```

```

        source: "../assets/loadcell.png"
        fillMode: Image.PreserveAspectFit
    }
}

Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Load Cell"; color:
"#dce8f5"; font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Basınç sensörü
kalibrasyonu"; color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }

Rectangle {
    anchors.horizontalCenter: parent.horizontalCenter
    visible: loadCellKayitliVar
    width: kaliMetni2.implicitWidth + 16
    height: 20
    radius: 10
    color: "#123321"

    Text {
        id: kaliMetni2
        anchors.centerIn: parent
        text: "✓ Kalibre Edildi"
        color: "#4ade80"
        font.pixelSize: 10
        font.bold: true
    }
}

}

MouseArea {
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onEntered: loadCellKarti.hovered = true
    onExited: loadCellKarti.hovered = false
    onClicked: seciliSensor = 1
}

}

Rectangle {
    id: mesafeKarti
    width: 350
    height: 450
    radius: 12
    color: "#0f1420"
    border.color: mesafeKarti.hovered ? "#9333ea" : "#1e2a3f"
    border.width: 1
    property bool hovered: false
    scale: hovered ? 1.03 : 1.0
    Behavior on border.color { ColorAnimation { duration: 120 } }
    Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }

    Column {
        anchors.centerIn: parent
        spacing: 14

        Item {
            anchors.horizontalCenter: parent.horizontalCenter
            width: 210
            height: 150

```

```

        Image {
            anchors.centerIn: parent
            width: parent.width
            height: parent.height
            source: "../assets/mesafe_sensor.png"
            fillMode: Image.PreserveAspectFit
        }
    }

    Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Mesafe Sensörü";
color: "#dce8f5"; font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
    Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Konum ölçümü
kalibrasyonu"; color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }

    Rectangle {
        anchors.horizontalCenter: parent.horizontalCenter
        visible: mesafeUstKayitliVar && mesafeAltKayitliVar
        width: kaliMetni3.implicitWidth + 16
        height: 20
        radius: 10
        color: "#123321"

        Text {
            id: kaliMetni3
            anchors.centerIn: parent
            text: "✓ Kalibre Edildi"
            color: "#4ade80"
            font.pixelSize: 10
            font.bold: true
        }
    }
}

MouseArea {
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onEntered: mesafeKarti.hovered = true
    onExited: mesafeKarti.hovered = false
    onClicked: { seciliSensor = 2; mesafeAdimi = 0 }
}

}

}

Item {
    Row {
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.margins: 20
        spacing: 12

        Button {
            text: "← Geri"
            font.pixelSize: 13

            background: Rectangle {
                radius: 6
                color: parent.hovered ? "#1e2a3f" : "#0f1420"
            }
        }
    }
}

```

```

        border.color: "#1e2a3f"
        border.width: 1
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#dce8f5"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
        leftPadding: 10
        rightPadding: 10
    }

    onClicked: seciliSensor = -1
}

Rectangle {
    width: 8
    height: 8
    radius: 4
    anchors.verticalCenter: parent.verticalCenter
    color: {
        if (seciliSensor === 0) return "#14b8a6"
        if (seciliSensor === 1) return "#3b82f6"
        if (seciliSensor === 2) return "#9333ea"
        return "#6b7280"
    }
}

Text {
    anchors.verticalCenter: parent.verticalCenter
    text: {
        if (seciliSensor === 0) return "Eğim Sensörü Kalibrasyonu"
        if (seciliSensor === 1) return "Load Cell Kalibrasyonu"
        if (seciliSensor === 2) return "Mesafe Sensörü Kalibrasyonu"
        return ""
    }
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 16
    font.bold: true
}

Item {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 70
    visible: seciliSensor === 0

    Column {
        anchors.centerIn: parent
        spacing: 20
        width: 360

        Rectangle {

```

```

anchors.horizontalCenter: parent.horizontalCenter
visible: egimKayitliVar
width: sonKaliMetni1.implicitWidth + 20
height: 26
radius: 13
color: "#0f1420"
border.color: "#1e2a3f"
border.width: 1

Text {
    id: sonKaliMetni1
    anchors.centerIn: parent
    text: "Son kalibrasyon: " + egimSonTarih
    color: "#6b7280"
    font.pixelSize: 11
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Cihazı düz ve sabit bir yüzeye yerleştirin"
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 14
}

SuTerazisi {
    anchors.horizontalCenter: parent.horizontalCenter
    egimX: sensorManager.egimX
    egimY: sensorManager.egimY
}

Rectangle {
    anchors.horizontalCenter: parent.horizontalCenter
    width: 220
    height: 72
    radius: 12
    color: "#0f2e2a"
    border.color: "#14b8a6"
    border.width: 1

    Text {
        anchors.centerIn: parent
        text: "X: " + sensorManager.egimX.toFixed(1) + "°   Y: " +
sensorManager.egimY.toFixed(1) + "°"
        color: "#2dd4bf"
        font.family: "Segoe UI"
        font.pixelSize: 22
        font.bold: true
    }
}

Button {
    width: parent.width
    height: 44
    text: "Sıfırla (Zero)"
    font.pixelSize: 14
    font.bold: true
}

```

```

        onClicked: {
            var eskiKal = database.kalibrasyonGetir("egim")
            var eskiOffsetX = eskiKal.mevcut ? eskiKal.deger1 : 0.0
            var eskiOffsetY = eskiKal.mevcut ? eskiKal.deger2 : 0.0
            var yeniOffsetX = eskiOffsetX + sensorManager.egimX
            var yeniOffsetY = eskiOffsetY + sensorManager.egimY
            database.kalibrasyonKaydet("egim", yeniOffsetX, yeniOffsetY)
            console.log("Egim sensoru kalibrasyonu kaydedildi.")
            seciliSensor = -1
        }

        background: Rectangle {
            radius: 8
            color: parent.hovered ? "#4f8cf7" : "#3b82f6"
            Behavior on color { ColorAnimation { duration: 120 } }
        }

        contentItem: Text {
            text: parent.text
            color: "#ffffff"
            font: parent.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }
}

```

```

Item {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 70
    visible: seciliSensor === 1

    Column {
        anchors.centerIn: parent
        spacing: 24
        width: 420
        visible: loadCellAraEkranGoster

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter
            width: araEkranBadge.implicitWidth + 20
            height: 26
            radius: 13
            color: "#0f1420"
            border.color: "#1e2a3f"
            border.width: 1

            Text {
                id: araEkranBadge
                anchors.centerIn: parent
                text: "Kayıtlı: " + loadCellNoktaSayisi + " nokta"
                color: "#6b7280"
                font.pixelSize: 11
            }
        }
    }
}

```

```

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Bu sensör için kayıtlı bir kalibrasyon bulundu"
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 16
    font.bold: true
}

Row {
    anchors.horizontalCenter: parent.horizontalCenter
    spacing: 20

    Rectangle {
        id: goruntuleKarti
        width: 190
        height: 180
        radius: 14
        color: "#0f1420"
        border.color: goruntuleKarti.hovered ? "#3b82f6" : "#1e2a3f"
        border.width: 1
        property bool hovered: false
        Behavior on border.color { ColorAnimation { duration: 120 } }

        Column {
            anchors.centerIn: parent
            spacing: 12
            width: parent.width - 28

            Canvas {
                anchors.horizontalCenter: parent.horizontalCenter
                width: 32
                height: 32
                onPaint: {
                    var ctx = getContext("2d")
                    ctx.clearRect(0, 0, width, height)
                    ctx.lineWidth = 1.8
                    ctx.lineCap = "round"
                    ctx.lineJoin = "round"
                    ctx.strokeStyle = "#3b82f6"

                    ctx.beginPath()
                    ctx.moveTo(3, 16)
                    ctx.bezierCurveTo(7.5, 8.5, 12.5, 6.8, 16, 6.8)
                    ctx.bezierCurveTo(19.5, 6.8, 24.5, 8.5, 29, 16)
                    ctx.bezierCurveTo(24.5, 23.5, 19.5, 25.2, 16, 25.2)
                    ctx.bezierCurveTo(12.5, 25.2, 7.5, 23.5, 3, 16)
                    ctx.stroke()

                    ctx.beginPath()
                    ctx.arc(16, 16, 4.4, 0, Math.PI * 2, false)
                    ctx.stroke()
                }
            }

            Text {
                width: parent.width

```

```

        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Mevcut Kalibrasyonu Görüntüle"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Kayıtlı noktaları incele"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 11
    }
}

MouseArea {
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onEntered: goruntuleKarti.hovered = true
    onExited: goruntuleKarti.hovered = false
    onClicked: {
        loadCellNoktalar = database.loadCellNoktalarGetir()
        loadCellAraEkranGoster = false
        loadCellGoruntuleModu = true
        loadCellDuzenlemeIndex = -1
    }
}

}

Rectangle {
    id: yenidenKarti
    width: 190
    height: 180
    radius: 14
    color: "#0f1420"
    border.color: yenidenKarti.hovered ? "#9ca3af" : "#1e2a3f"
    border.width: 1
    property bool hovered: false
    Behavior on border.color { ColorAnimation { duration: 120 } }

    Column {
        anchors.centerIn: parent
        spacing: 12
        width: parent.width - 28

        Canvas {
            anchors.horizontalCenter: parent.horizontalCenter
            width: 32
            height: 32
            onPaint: {
                var ctx = getContext("2d")
                ctx.clearRect(0, 0, width, height)
                ctx.strokeStyle = "#9ca3af"
                ctx.fillStyle = "#9ca3af"
            }
        }
    }
}

```

```

ctx.lineWidth = 3.4
ctx.lineCap = "butt"

function yayOku(startDeg, endDeg) {
    var s = startDeg * Math.PI / 180
    var e = endDeg * Math.PI / 180
    var cx = 16, cy = 16, r = 10.5

    ctx.beginPath()
    ctx.arc(cx, cy, r, s, e, false)
    ctx.stroke()

    var ex = cx + r * Math.cos(e)
    var ey = cy + r * Math.sin(e)
    var tx = -Math.sin(e)
    var ty = Math.cos(e)
    var px = -ty
    var py = tx
    var geri = 6.2
    var yaricap = 3.6
    var bx = ex - tx * geri
    var by = ey - ty * geri

    ctx.beginPath()
    ctx.moveTo(ex, ey)
    ctx.lineTo(bx + px * yaricap, by + py * yaricap)
    ctx.lineTo(bx - px * yaricap, by - py * yaricap)
    ctx.closePath()
    ctx.fill()
}

yayOku(195, 345)
yayOku(15, 165)
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Yeniden Kalibre Et"
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Sıfırdan yeni ölçüm al"
    color: "#6b7280"
    font.family: "Segoe UI"
    font.pixelSize: 11
}

MouseArea {
    anchors.fill: parent

```

```

        hoverEnabled: true
        cursorShape: Qt.PointingHandCursor
        onEntered: yenidenKarti.hovered = true
        onExited: yenidenKarti.hovered = false
        onClicked: loadCellYenidenOnayGoster = true
    }
}
}
}

Column {
    anchors.centerIn: parent
    spacing: 16
    width: 340
    visible: loadCellGoruntuleModu

    Text {
        text: "KAYITLI KALİBRASYON NOKTALARI"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 11
        font.bold: true
        font.letterSpacing: 1
    }

    Rectangle {
        width: parent.width
        height: 320
        radius: 14
        color: "#0f1420"
        border.color: "#1e2a3f"
        border.width: 1

        ListView {
            anchors.fill: parent
            anchors.margins: 16
            clip: true
            spacing: 8

            model: loadCellNoktalari

            delegate: Rectangle {
                width: ListView.view.width
                height: 48
                radius: 8
                color: "#0a0e17"
                border.color: "#1e2a3f"
                border.width: 1

                Row {
                    anchors.fill: parent
                    anchors.leftMargin: 12
                    anchors.rightMargin: 12
                    spacing: 8

                    Rectangle {
                        width: 22
                        height: 22
                        radius: 11
                        anchors.verticalCenter: parent.verticalCenter

```

```

        color: "#182644"
        border.color: "#3b82f6"
        border.width: 1

        Text {
            anchors.centerIn: parent
            text: index + 1
            color: "#3b82f6"
            font.pixelSize: 10
            font.bold: true
        }
    }

    Column {
        anchors.verticalCenter: parent.verticalCenter
        spacing: 1

        Text {
            text: modelData.hedefKg.toFixed(2) + " kg"
            color: "#dce8f5"
            font.family: "Segoe UI"
            font.pixelSize: 13
            font.bold: true
        }

        Text {
            text: "ADC: " + modelData.hamDeger.toFixed(0)
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
        }
    }
}

Button {
    width: parent.width
    height: 46
    text: "Düzenle"
    font.pixelSize: 14
    font.bold: true

    onClicked: {
        loadCellToplamNokta = loadCellNoktalari.length
        loadCellGoruntuleModu = false
        loadCellTamDuzenlemeModu = true
        loadCellSayiSecildi = true
        loadCellDuzenlemeIndex = -1
    }

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#4f8cf7" : "#3b82f6"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
    }
}

```

```

        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}
}

Rectangle {
    anchors.fill: parent
    color: "#000000b3"
    visible: loadCellYenidenOnayGoster
    z: 200

    MouseArea { anchors.fill: parent }

    Rectangle {
        anchors.centerIn: parent
        width: 360
        height: 200
        radius: 14
        color: "#0f1420"
        border.color: "#1e2a3f"
        border.width: 1

        Column {
            anchors.centerIn: parent
            spacing: 18
            width: parent.width - 40

            Text {
                width: parent.width
                horizontalAlignment: Text.AlignHCenter
                wrapMode: Text.WordWrap
                text: "Mevcut kalibrasyon silinecek, emin misiniz?"
                color: "#dce8f5"
                font.family: "Segoe UI"
                font.pixelSize: 14
                font.bold: true
            }

            Row {
                anchors.horizontalCenter: parent.horizontalCenter
                spacing: 12

                Button {
                    width: 150
                    height: 42
                    text: "Vazgeç"

                    onClicked: loadCellYenidenOnayGoster = false

                    background: Rectangle {
                        radius: 8
                        color: parent.hovered ? "#1e2a3f" : "transparent"
                        border.color: "#1e2a3f"
                        border.width: 1
                        Behavior on color { ColorAnimation { duration: 120 } }
                    }
                }
            }
        }
    }
}

```

```

        contentItem: Text {
            text: parent.text
            color: "#9ca3af"
            font: parent.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }

    Button {
        width: 150
        height: 42
        text: "Evet, Yeniden Başla"
        font.bold: true

        onClicked: {
            loadCellYenidenOnayGoster = false
            loadCellAraEkranGoster = false
            loadCellGoruntuleModu = false
            loadCellTamDuzenlemeModu = false
            loadCellNoktalari = []
            loadCellSayiSecildi = false
            loadCellDuzenlemeIndex = -1
        }

        background: Rectangle {
            radius: 8
            color: parent.hovered ? "#b91c1c" : "#991b1b"
            Behavior on color { ColorAnimation { duration: 120 } }
        }

        contentItem: Text {
            text: parent.text
            color: "#ffffff"
            font: parent.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }
}

}

}

}

Rectangle {
    anchors.fill: parent
    color: "#000000b3"
    visible: loadCellTamamlandiOnayGoster
    z: 200

    MouseArea { anchors.fill: parent }

    Rectangle {
        anchors.centerIn: parent
        width: 380
        height: 220
        radius: 14
        color: "#0f1420"
        border.color: "#16a34a"
        border.width: 1
    }

```

```

Column {
    anchors.centerIn: parent
    spacing: 18
    width: parent.width - 40

    Column {
        width: parent.width
        spacing: 6

        Text {
            width: parent.width
            horizontalAlignment: Text.AlignHCenter
            text: "Kalibrasyon Tamamlandı"
            color: "#4ade80"
            font.family: "Segoe UI"
            font.pixelSize: 17
            font.bold: true
        }

        Text {
            width: parent.width
            horizontalAlignment: Text.AlignHCenter
            wrapMode: Text.WordWrap
            text: "Toplam " + loadCellToplamNokta + " nokta girildi. Bu kalibrasyonu
kaydetmek istiyor musunuz?"

            color: "#9ca3af"
            font.family: "Segoe UI"
            font.pixelSize: 13
        }
    }

    Row {
        anchors.horizontalCenter: parent.horizontalCenter
        spacing: 12

        Button {
            width: 150
            height: 42
            text: "Noktaları Kontrol Et"

            onClicked: loadCellTamamlandiOnayGoster = false

            background: Rectangle {
                radius: 8
                color: parent.hovered ? "#1e2a3f" : "transparent"
                border.color: "#1e2a3f"
                border.width: 1
                Behavior on color { ColorAnimation { duration: 120 } }
            }

            contentItem: Text {
                text: parent.text
                color: "#9ca3af"
                font: parent.font
                horizontalAlignment: Text.AlignHCenter
                verticalAlignment: Text.AlignVCenter
            }
        }
    }
}

```

```

        Button {
            width: 150
            height: 42
            text: "Kaydet ve Bitir"
            font.bold: true

            onClicked: {
                loadCellTamamlandiOnayGoster = false
                loadCellKaydetVeBildir()
            }

            background: Rectangle {
                radius: 8
                color: parent.hovered ? "#22c55e" : "#16a34a"
                Behavior on color { ColorAnimation { duration: 120 } }
            }

            contentItem: Text {
                text: parent.text
                color: "#ffffff"
                font: parent.font
                horizontalAlignment: Text.AlignHCenter
                verticalAlignment: Text.AlignVCenter
            }
        }
    }
}

```

```

Rectangle {
    id: loadCellBildirimKutusu
    anchors.top: parent.top
    anchors.horizontalCenter: parent.horizontalCenter
    anchors.topMargin: 12
    z: 100
    width: loadCellBildirimMetni.implicitWidth + 24
    height: 32
    radius: 16
    color: loadCellBildirimBasarili ? "#123321" : "#3a1414"
    opacity: loadCellBildirimGoster ? 1 : 0
    visible: opacity > 0
    Behavior on opacity { NumberAnimation { duration: 200 } }

    Text {
        id: loadCellBildirimMetni
        anchors.centerIn: parent
        text: loadCellBildirimMesaj
        color: loadCellBildirimBasarili ? "#4ade80" : "#f87171"
        font.family: "Segoe UI"
        font.pixelSize: 12
        font.bold: true
    }
}

```

```

Column {
    anchors.centerIn: parent
    spacing: 24
    width: 360
    visible: !loadCellSayiSecildi && !loadCellAraEkranGoster && !loadCellGoruntuleModu
}

```

```

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Kaç noktalı kalibrasyon yapmak istersiniz?"
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 16
    font.bold: true
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Daha fazla nokta, daha hassas kalibrasyon sağlar"
    color: "#6b7280"
    font.family: "Segoe UI"
    font.pixelSize: 12
}

Grid {
    anchors.horizontalCenter: parent.horizontalCenter
    columns: 4
    spacing: 12

    Repeater {
        model: [4, 6, 8, 10]

        Rectangle {
            id: secKarti
            width: 72
            height: 72
            radius: 12
            color: secKarti.hovered ? "#182644" : "#0f1420"
            border.color: "#3b82f6"
            border.width: 1
            property bool hovered: false
            Behavior on color { ColorAnimation { duration: 120 } }

            Text {
                anchors.centerIn: parent
                text: modelData
                color: "#dce8f5"
                font.family: "Segoe UI"
                font.pixelSize: 26
                font.bold: true
            }

            MouseArea {
                anchors.fill: parent
                hoverEnabled: true
                cursorShape: Qt.PointingHandCursor
                onEntered: secKarti.hovered = true
                onExited: secKarti.hovered = false
                onClicked: {
                    loadCellToplamNokta = modelData
                    loadCellNoktalari = []
                    loadCellDuzenlemeIndex = -1
                }
            }
        }
    }
}

```

```

        loadCellSayiSecildi = true
    }
}
}
}
}
}
}

Row {
    anchors.centerIn: parent
    spacing: 24
    visible: loadCellSayiSecildi

    Rectangle {
        width: 340
        height: 500
        radius: 14
        color: "#0f1420"
        border.color: "#1e2a3f"
        border.width: 1

        Column {
            anchors.centerIn: parent
            spacing: 16
            width: parent.width - 40

            Rectangle {
                anchors.horizontalCenter: parent.horizontalCenter
                visible: loadCellKayitliVar && !loadCellTamDuzenlemeModu
                width: sonKaliMetni2.implicitWidth + 20
                height: 24
                radius: 12
                color: "#0a0e17"
                border.color: "#1e2a3f"
                border.width: 1

                Text {
                    id: sonKaliMetni2
                    anchors.centerIn: parent
                    text: "Kayıtlı: " + loadCellNoktaSayisi + " nokta"
                    color: "#6b7280"
                    font.pixelSize: 11
                }
            }

            Text {
                anchors.horizontalCenter: parent.horizontalCenter
                text: loadCellTamDuzenlemeModu
                    ? "Kalibrasyonu Düzenle"
                    : "Nokta " + (loadCellNoktalari.length + 1) + " / " +

loadCellToplamNokta

                color: "#3b82f6"
                font.family: "Segoe UI"
                font.pixelSize: 13
                font.bold: true
                font.letterSpacing: 1
            }

            Rectangle {
                width: parent.width

```

```

height: 6
radius: 3
color: "#1e2a3f"

Rectangle {
    width: parent.width * (loadCellNoktalari.length / loadCellToplamNokta)
    height: parent.height
    radius: 3
    color: "#3b82f6"
    Behavior on width { NumberAnimation { duration: 150 } }
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: loadCellDuzenlemeIndex >= 0
        ? "Nokta " + (loadCellDuzenlemeIndex + 1) + " düzenleniyor"
        : (loadCellTamDuzenlemeModu
            ? "Düzenlemek istediğiniz noktaya sağdaki listeden tıklayın"
            : "Hedef ağırlığı gir, sonra o ağırlığı load cell'e koy")
    color: loadCellDuzenlemeIndex >= 0 ? "#3b82f6" : "#9ca3af"
    font.family: "Segoe UI"
    font.bold: loadCellDuzenlemeIndex >= 0
    font.pixelSize: 13
}

TextField {
    id: hedefKgKutusu
    anchors.horizontalCenter: parent.horizontalCenter
    width: 180
    height: 44
    placeholderText: "kg"
    placeholderTextColor: "#4b5563"
    color: "ffffff"
    font.pixelSize: 16
    horizontalAlignment: TextInput.AlignHCenter
    validator: DoubleValidator { bottom: 0; top: 500; decimals: 2 }
    background: Rectangle {
        color: "#0a0e17"
        radius: 10
        border.color: hedefKgKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
        border.width: 1
    }
}

Rectangle {
    anchors.horizontalCenter: parent.horizontalCenter
    width: parent.width
    height: 64
    radius: 12
    color: "#132335"
    border.color: wifiManager.baglandi ? "#3b82f6" : "#dc2626"
    border.width: 1
    Behavior on border.color { ColorAnimation { duration: 120 } }

    Text {
        anchors.centerIn: parent
        text: "Ham değer: " + sensorManager.hamAgirlik.toFixed(0)
    }
}

```

```

        color: wifiManager.baglandi ? "#3b82f6" : "#f87171"
        font.family: "Segoe UI"
        font.pixelSize: 20
        font.bold: true
    }
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    visible: !wifiManager.baglandi
    text: "Cihaz bağlı değil – ham değer 0 olarak kaydedilecek"
    color: "#f87171"
    font.family: "Segoe UI"
    font.pixelSize: 11
}

Button {
    id: noktaKaydetButonu
    width: parent.width
    height: 46
    property bool tumNoktalarGirildi: !loadCellTamDuzenlemeModu &&
loadCellDuzenlemeIndex < 0 && loadCellNoktalari.length >= loadCellToplamNokta
    text: loadCellDuzenlemeIndex >= 0
        ? "Noktayı Güncelle"
        : (tumNoktalarGirildi
            ? "Kalibrasyonu Tamamla"
            : (loadCellNoktalari.length >= loadCellToplamNokta - 1 ? "Son
Noktayı Kaydet" : "Bu Noktayı Kaydet"))
    font.pixelSize: 14
    font.bold: true
    enabled: tumNoktalarGirildi || (hedefKgKutusu.text.length > 0 &&
(loadCellDuzenlemeIndex >= 0 || loadCellNoktalari.length < loadCellToplamNokta))

    onClicked: {
        if (tumNoktalarGirildi) {
            loadCellTamamlandiOnayGoster = true
            return
        }

        var yeniNokta = { hedefKg: parseFloat(hedefKgKutusu.text), hamDeger:

sensorManager.hamAgirlik }

        if (loadCellDuzenlemeIndex >= 0) {
            // Var olan bir noktayı yerinde güncelliyoruz, diğer noktalar

            var guncelDizi = loadCellNoktalari.slice()
            guncelDizi[loadCellDuzenlemeIndex] = yeniNokta
            loadCellNoktalari = guncelDizi
            loadCellDuzenlemeIndex = -1
            hedefKgKutusu.text = ""
            return
        }

        var yeniDizi = loadCellNoktalari.slice()
        yeniDizi.push(yeniNokta)
        loadCellNoktalari = yeniDizi
        hedefKgKutusu.text = ""
    }
}

```

"#1e2a3f"

```
        if (loadCellNoktalari.length >= loadCellToplamNokta) {
            loadCellTamamlandiOnayGoster = true
        }
    }

    background: Rectangle {
        radius: 10
        color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") :

        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Button {
    width: parent.width
    height: 46
    visible: loadCellTamDuzenlemeModu
    text: "Değişiklikleri Kaydet"
    font.pixelSize: 14
    font.bold: true

    onClicked: loadCellKaydetVeBildir()

    background: Rectangle {
        radius: 10
        color: parent.hovered ? "#22c55e" : "#16a34a"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Button {
    width: parent.width
    height: 38
    visible: loadCellDuzenlemeIndex >= 0
    text: "Vazgeç"
    font.pixelSize: 13

    onClicked: {
        loadCellDuzenlemeIndex = -1
        hedefKgKutusu.text = ""
    }

    background: Rectangle {
        radius: 8
```



```

delegate: Rectangle {
    width: ListView.view.width
    height: 48
    radius: 8
    color: satirAlani.containsMouse ? "#182644" : "#0a0e17"
    border.color: index === loadCellDuzenlemeIndex ? "#3b82f6" : "#1e2a3f"
    border.width: index === loadCellDuzenlemeIndex ? 2 : 1
    Behavior on color { ColorAnimation { duration: 120 } }

    Row {
        anchors.fill: parent
        anchors.leftMargin: 12
        anchors.rightMargin: 12
        spacing: 8

        Rectangle {
            width: 22
            height: 22
            radius: 11
            anchors.verticalCenter: parent.verticalCenter
            color: "#182644"
            border.color: "#3b82f6"
            border.width: 1

            Text {
                anchors.centerIn: parent
                text: index + 1
                color: "#3b82f6"
                font.pixelSize: 10
                font.bold: true
            }
        }

        Column {
            anchors.verticalCenter: parent.verticalCenter
            spacing: 1

            Text {
                text: modelData.hedefKg.toFixed(2) + " kg"
                color: "#dce8f5"
                font.family: "Segoe UI"
                font.pixelSize: 13
                font.bold: true
            }

            Text {
                text: "ADC: " + modelData.hamDeger.toFixed(0)
                color: "#6b7280"
                font.family: "Segoe UI"
                font.pixelSize: 10
            }
        }
    }

    Text {
        anchors.right: parent.right
        anchors.verticalCenter: parent.verticalCenter
        anchors.rightMargin: 12
        text: "↩"
    }
}

```

```

        color: "#6b7280"
        font.pixelSize: 13
        visible: satirAlani.containsMouse
    }

    MouseArea {
        id: satirAlani
        anchors.fill: parent
        hoverEnabled: true
        cursorShape: Qt.PointingHandCursor
        onClicked: {
            hedefKgKutusu.text = modelData.hedefKg.toString()
            loadCellDuzenlemeIndex = index
        }
    }
}

Item {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 70
    visible: seciliSensor === 2

    Text {
        anchors.top: parent.top
        anchors.horizontalCenter: parent.horizontalCenter
        text: mesafeAsama === 0 ? "AŞAMA 1/2 – ÖLÇÜM KALİBRASYONU" : "AŞAMA 2/2 – STROKE"

        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 11
        font.bold: true
        font.letterSpacing: 1
    }
}

// =====
// A) OLCUM KALIBRASYONU – cok nokta (load cell birebir kopyasi)
// =====
Item {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 30
    visible: mesafeAsama === 0

    Column {
        anchors.centerIn: parent
        spacing: 24
        width: 420
        visible: mesafeAraEkranGoster

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter

```

```

width: mesafeAraEkranBadge.implicitWidth + 20
height: 26
radius: 13
color: "#0f1420"
border.color: "#1e2a3f"
border.width: 1

Text {
    id: mesafeAraEkranBadge
    anchors.centerIn: parent
    text: "Kayıtlı: " + mesafeOlcumNoktaSayisi + " nokta"
    color: "#6b7280"
    font.pixelSize: 11
}

}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Bu sensör için kayıtlı bir kalibrasyon bulundu"
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 16
    font.bold: true
}

Row {
    anchors.horizontalCenter: parent.horizontalCenter
    spacing: 20

    Rectangle {
        id: mesafeGoruntuleKarti
        width: 190
        height: 180
        radius: 14
        color: "#0f1420"
        border.color: mesafeGoruntuleKarti.hovered ? "#9333ea" : "#1e2a3f"
        border.width: 1
        property bool hovered: false
        Behavior on border.color { ColorAnimation { duration: 120 } }

        Column {
            anchors.centerIn: parent
            spacing: 12
            width: parent.width - 28

            Canvas {
                anchors.horizontalCenter: parent.horizontalCenter
                width: 32
                height: 32
                onPaint: {
                    var ctx = getContext("2d")
                    ctx.clearRect(0, 0, width, height)
                    ctx.lineWidth = 1.8
                    ctx.lineCap = "round"
                    ctx.lineJoin = "round"
                    ctx.strokeStyle = "#9333ea"

                    ctx.beginPath()

```

```

        ctx.moveTo(3, 16)
        ctx.bezierCurveTo(7.5, 8.5, 12.5, 6.8, 16, 6.8)
        ctx.bezierCurveTo(19.5, 6.8, 24.5, 8.5, 29, 16)
        ctx.bezierCurveTo(24.5, 23.5, 19.5, 25.2, 16, 25.2)
        ctx.bezierCurveTo(12.5, 25.2, 7.5, 23.5, 3, 16)
        ctx.stroke()

        ctx.beginPath()
        ctx.arc(16, 16, 4.4, 0, Math.PI * 2, false)
        ctx.stroke()
    }
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Mevcut Kalibrasyonu Görüntüle"
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Kayıtlı noktaları incele"
    color: "#6b7280"
    font.family: "Segoe UI"
    font.pixelSize: 11
}

}

MouseArea {
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onEntered: mesafeGoruntuleKarti.hovered = true
    onExited: mesafeGoruntuleKarti.hovered = false
    onClicked: {
        mesafeNoktaları = database.mesafeNoktalarıGetir()
        mesafeAraEkranGoster = false
        mesafeGoruntuleModu = true
        mesafeDuzenlemeIndex = -1
    }
}

}

Rectangle {
    id: mesafeYenidenKarti
    width: 190
    height: 180
    radius: 14
    color: "#0f1420"
    border.color: mesafeYenidenKarti.hovered ? "#9ca3af" : "#1e2a3f"
    border.width: 1
    property bool hovered: false
    Behavior on border.color { ColorAnimation { duration: 120 } }

```

```

Column {
    anchors.centerIn: parent
    spacing: 12
    width: parent.width - 28

    Canvas {
        anchors.horizontalCenter: parent.horizontalCenter
        width: 32
        height: 32
        onPaint: {
            var ctx = getContext("2d")
            ctx.clearRect(0, 0, width, height)
            ctx.strokeStyle = "#9ca3af"
            ctx.fillStyle = "#9ca3af"
            ctx.lineWidth = 3.4
            ctx.lineCap = "butt"

            function yayOku(startDeg, endDeg) {
                var s = startDeg * Math.PI / 180
                var e = endDeg * Math.PI / 180
                var cx = 16, cy = 16, r = 10.5

                ctx.beginPath()
                ctx.arc(cx, cy, r, s, e, false)
                ctx.stroke()

                var ex = cx + r * Math.cos(e)
                var ey = cy + r * Math.sin(e)
                var tx = -Math.sin(e)
                var ty = Math.cos(e)
                var px = -ty
                var py = tx
                var geri = 6.2
                var yaricap = 3.6
                var bx = ex - tx * geri
                var by = ey - ty * geri

                ctx.beginPath()
                ctx.moveTo(ex, ey)
                ctx.lineTo(bx + px * yaricap, by + py * yaricap)
                ctx.lineTo(bx - px * yaricap, by - py * yaricap)
                ctx.closePath()
                ctx.fill()
            }

            yayOku(195, 345)
            yayOku(15, 165)
        }
    }

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Yeniden Kalibre Et"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }
}

```

```

    }

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Sıfırdan yeni ölçüm al"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 11
    }
}

MouseArea {
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onEntered: mesafeYenidenKarti.hovered = true
    onExited: mesafeYenidenKarti.hovered = false
    onClicked: mesafeYenidenOnayGoster = true
}
}
}
}
}

```

```

Column {
    anchors.centerIn: parent
    spacing: 16
    width: 340
    visible: mesafeGoruntuleModu

    Text {
        text: "KAYITLI KALİBRASYON NOKTALARI"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 11
        font.bold: true
        font.letterSpacing: 1
    }
}

```

```

Rectangle {
    width: parent.width
    height: 320
    radius: 14
    color: "#0f1420"
    border.color: "#1e2a3f"
    border.width: 1

    ListView {
        anchors.fill: parent
        anchors.margins: 16
        clip: true
        spacing: 8

        model: mesafeNoktalari

        delegate: Rectangle {
            width: ListView.view.width
            height: 48
            radius: 8
        }
    }
}

```

```
color: "#0a0e17"
border.color: "#1e2a3f"
border.width: 1
```

```
Row {
    anchors.fill: parent
    anchors.leftMargin: 12
    anchors.rightMargin: 12
    spacing: 8
```

```
    Rectangle {
        width: 22
        height: 22
        radius: 11
        anchors.verticalCenter: parent.verticalCenter
        color: "#241a38"
        border.color: "#9333ea"
        border.width: 1
```

```
        Text {
            anchors.centerIn: parent
            text: index + 1
            color: "#c084fc"
            font.pixelSize: 10
            font.bold: true
        }
    }
```

```
}
```

```
Column {
    anchors.verticalCenter: parent.verticalCenter
    spacing: 1
```

```
    Text {
        text: (modelData.hedefMm / 10).toFixed(1) + " cm"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }
```

```
    Text {
        text: "ADC: " + modelData.hamDeger.toFixed(0)
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 10
    }
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
Button {
    width: parent.width
    height: 46
    text: "Düzenle"
    font.pixelSize: 14
    font.bold: true

    onClicked: {
```

```

        mesafeToplamNokta = mesafeNoktalari.length
        mesafeGoruntuleModu = false
        mesafeTamDuzenlemeModu = true
        mesafeSayiSecildi = true
        mesafeDuzenlemeIndex = -1
    }

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#a855f7" : "#9333ea"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Rectangle {
    anchors.fill: parent
    color: "#000000b3"
    visible: mesafeYenidenOnayGoster
    z: 200

    MouseArea { anchors.fill: parent }

    Rectangle {
        anchors.centerIn: parent
        width: 360
        height: 200
        radius: 14
        color: "#0f1420"
        border.color: "#1e2a3f"
        border.width: 1

        Column {
            anchors.centerIn: parent
            spacing: 18
            width: parent.width - 40

            Text {
                width: parent.width
                horizontalAlignment: Text.AlignHCenter
                wrapMode: Text.WordWrap
                text: "Mevcut kalibrasyon silinecek, emin misiniz?"
                color: "#dce8f5"
                font.family: "Segoe UI"
                font.pixelSize: 14
                font.bold: true
            }

            Row {
                anchors.horizontalCenter: parent.horizontalCenter
                spacing: 12
            }
        }
    }
}

```

```

Button {
    width: 150
    height: 42
    text: "Vazgeç"

    onClicked: mesafeYenidenOnayGoster = false

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#1e2a3f" : "transparent"
        border.color: "#1e2a3f"
        border.width: 1
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#9ca3af"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

```

```

Button {
    width: 150
    height: 42
    text: "Evet, Yeniden Başla"
    font.bold: true

    onClicked: {
        mesafeYenidenOnayGoster = false
        mesafeAraEkranGoster = false
        mesafeGoruntuleModu = false
        mesafeTamDuzenlemeModu = false
        mesafeNoktalarlari = []
        mesafeSayiSecildi = false
        mesafeDuzenlemeIndex = -1
    }

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#b91c1c" : "#991b1b"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

```

```

}
}
}
}
}
Rectangle {

```

```
anchors.fill: parent
color: "#000000b3"
visible: mesafeTamamlandiOnayGoster
z: 200
```

```
MouseArea { anchors.fill: parent }
```

```
Rectangle {
    anchors.centerIn: parent
    width: 380
    height: 220
    radius: 14
    color: "#0f1420"
    border.color: "#16a34a"
    border.width: 1
```

```
Column {
    anchors.centerIn: parent
    spacing: 18
    width: parent.width - 40
```

```
Column {
    width: parent.width
    spacing: 6
```

```
Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    text: "Kalibrasyon Tamamlandı"
    color: "#4ade80"
    font.family: "Segoe UI"
    font.pixelSize: 17
    font.bold: true
}
```

```
Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: "Toplam " + mesafeToplamNokta + " nokta girildi. Bu
kalibrasyonu kaydetmek istiyor musunuz?"
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 13
}
```

```
Row {
    anchors.horizontalCenter: parent.horizontalCenter
    spacing: 12
```

```
Button {
    width: 150
    height: 42
    text: "Noktaları Kontrol Et"

    onClicked: mesafeTamamlandiOnayGoster = false

    background: Rectangle {
        radius: 8
```

```

        color: parent.hovered ? "#1e2a3f" : "transparent"
        border.color: "#1e2a3f"
        border.width: 1
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#9ca3af"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Button {
    width: 150
    height: 42
    text: "Kaydet ve Bitir"
    font.bold: true

    onClicked: {
        mesafeTamamlandiOnayGoster = false
        mesafeKaydetVeBildir()
    }

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#22c55e" : "#16a34a"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}
}
}
}

Column {
    anchors.centerIn: parent
    spacing: 24
    width: 360
    visible: !mesafeSayiSecildi && !mesafeAraEkranGoster && !mesafeGoruntuleModu

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Kaç noktalı kalibrasyon yapmak istersiniz?"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 16
        font.bold: true
    }
}

```

```

    }

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Daha fazla nokta, daha hassas kalibrasyon sağlar"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 12
    }

    Grid {
        anchors.horizontalCenter: parent.horizontalCenter
        columns: 4
        spacing: 12

        Repeater {
            model: [4, 6, 8, 10]

            Rectangle {
                id: mesafeSecKarti
                width: 72
                height: 72
                radius: 12
                color: mesafeSecKarti.hovered ? "#241a38" : "#0f1420"
                border.color: "#9333ea"
                border.width: 1
                property bool hovered: false
                Behavior on color { ColorAnimation { duration: 120 } }

                Text {
                    anchors.centerIn: parent
                    text: modelData
                    color: "#dce8f5"
                    font.family: "Segoe UI"
                    font.pixelSize: 26
                    font.bold: true
                }

                MouseArea {
                    anchors.fill: parent
                    hoverEnabled: true
                    cursorShape: Qt.PointingHandCursor
                    onEntered: mesafeSecKarti.hovered = true
                    onExited: mesafeSecKarti.hovered = false
                    onClicked: {
                        mesafeToplamNokta = modelData
                        mesafeNoktalari = []
                        mesafeDuzenlemeIndex = -1
                        mesafeSayiSecildi = true
                    }
                }
            }
        }
    }
}

Row {
    anchors.centerIn: parent

```

```
spacing: 24
visible: mesafeSayiSecildi
```

```
Rectangle {
    width: 340
    height: 500
    radius: 14
    color: "#0f1420"
    border.color: "#1e2a3f"
    border.width: 1

    Column {
        anchors.centerIn: parent
        spacing: 16
        width: parent.width - 40

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter
            visible: mesafeOlcumKayitliVar && !mesafeTamDuzenlemeModu
            width: mesafeSonKaliMetni.implicitWidth + 20
            height: 24
            radius: 12
            color: "#0a0e17"
            border.color: "#1e2a3f"
            border.width: 1

            Text {
                id: mesafeSonKaliMetni
                anchors.centerIn: parent
                text: "Kayıtlı: " + mesafeOlcumNoktaSayisi + " nokta"
                color: "#6b7280"
                font.pixelSize: 11
            }
        }
    }
}
```

mesafeToplamNokta

```
Text {
    anchors.horizontalCenter: parent.horizontalCenter
    text: mesafeTamDuzenlemeModu
        ? "Kalibrasyonu Düzenle"
        : "Nokta " + (mesafeNoktalari.length + 1) + " / " +

    color: "#9333ea"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
    font.letterSpacing: 1
}

Rectangle {
    width: parent.width
    height: 6
    radius: 3
    color: "#1e2a3f"

    Rectangle {
        width: parent.width * (mesafeNoktalari.length / mesafeToplamNokta)
        height: parent.height
        radius: 3
        color: "#9333ea"
        Behavior on width { NumberAnimation { duration: 150 } }
    }
}
```

```

    }
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: mesafeDuzenlemeIndex >= 0
        ? "Nokta " + (mesafeDuzenlemeIndex + 1) + " düzenleniyor"
        : (mesafeTamDuzenlemeModu
            ? "Düzenlemek istediğiniz noktaya sağdaki listeden tıklayın"
            : "Hedef mesafeyi gir, sonra boruyu o mesafeye getir")
    color: mesafeDuzenlemeIndex >= 0 ? "#9333ea" : "#9ca3af"
    font.family: "Segoe UI"
    font.bold: mesafeDuzenlemeIndex >= 0
    font.pixelSize: 13
}

TextField {
    id: hedefCmKutusu
    anchors.horizontalCenter: parent.horizontalCenter
    width: 180
    height: 44
    placeholderText: "cm"
    placeholderTextColor: "#4b5563"
    color: "#ffffff"
    font.pixelSize: 16
    horizontalAlignment: TextInput.AlignHCenter
    validator: DoubleValidator { bottom: 0; top: 300; decimals: 1 }
    background: Rectangle {
        color: "#0a0e17"
        radius: 10
        border.color: hedefCmKutusu.activeFocus ? "#9333ea" : "#1e2a3f"
        border.width: 1
    }
}

Rectangle {
    anchors.horizontalCenter: parent.horizontalCenter
    width: parent.width
    height: 64
    radius: 12
    color: "#132335"
    border.color: wifiManager.baglandi ? "#9333ea" : "#dc2626"
    border.width: 1
    Behavior on border.color { ColorAnimation { duration: 120 } }

    Text {
        anchors.centerIn: parent
        text: "Ham değer: " + sensorManager.hamMesafe.toFixed(0)
        color: wifiManager.baglandi ? "#c084fc" : "#f87171"
        font.family: "Segoe UI"
        font.pixelSize: 20
        font.bold: true
    }
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter

```

```

wrapMode: Text.WordWrap
visible: !wifiManager.baglandi
text: "Cihaz bağlı değil – ham değer 0 olarak kaydedilecek"
color: "#f87171"
font.family: "Segoe UI"
font.pixelSize: 11
}

Button {
    id: mesafeNoktaKaydetButonu
    width: parent.width
    height: 46
    property bool tumNoktalarGirildi: !mesafeTamDuzenlemeModu &&
mesafeDuzenlemeIndex < 0 && mesafeNoktalar.length >= mesafeToplamNokta
    text: mesafeDuzenlemeIndex >= 0
        ? "Noktayı Güncelle"
        : (tumNoktalarGirildi
            ? "Kalibrasyonu Tamamla"
            : (mesafeNoktalar.length >= mesafeToplamNokta - 1 ? "Son
Noktayı Kaydet" : "Bu Noktayı Kaydet"))
    font.pixelSize: 14
    font.bold: true
    enabled: tumNoktalarGirildi || (hedefCmKutusu.text.length > 0 &&
(mesafeDuzenlemeIndex >= 0 || mesafeNoktalar.length < mesafeToplamNokta))

    onClicked: {
        if (tumNoktalarGirildi) {
            mesafeTamamlandiOnayGoster = true
            return
        }

        var yeniNokta = { hedefMm: parseFloat(hedefCmKutusu.text) * 10,
hamDeger: sensorManager.hamMesafe }

        if (mesafeDuzenlemeIndex >= 0) {
            // Var olan bir noktayı yerinde güncelliyoruz, diğer noktalar
etkilenmiyor

            var guncelDizi = mesafeNoktalar.slice()
            guncelDizi[mesafeDuzenlemeIndex] = yeniNokta
            mesafeNoktalar = guncelDizi
            mesafeDuzenlemeIndex = -1
            hedefCmKutusu.text = ""
            return
        }

        var yeniDizi = mesafeNoktalar.slice()
        yeniDizi.push(yeniNokta)
        mesafeNoktalar = yeniDizi
        hedefCmKutusu.text = ""

        if (mesafeNoktalar.length >= mesafeToplamNokta) {
            mesafeTamamlandiOnayGoster = true
        }
    }

    background: Rectangle {
        radius: 10
        color: parent.enabled ? (parent.hovered ? "#a855f7" : "#9333ea") :
"#1e2a3f"

        Behavior on color { ColorAnimation { duration: 120 } }

```

```

    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Button {
    width: parent.width
    height: 46
    visible: mesafeTamDuzenlemeModu
    text: "Değişiklikleri Kaydet"
    font.pixelSize: 14
    font.bold: true

    onClicked: mesafeKaydetVeBildir()

    background: Rectangle {
        radius: 10
        color: parent.hovered ? "#22c55e" : "#16a34a"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Button {
    width: parent.width
    height: 38
    visible: mesafeDuzenlemeIndex >= 0
    text: "Vazgeç"
    font.pixelSize: 13

    onClicked: {
        mesafeDuzenlemeIndex = -1
        hedefCmKutusu.text = ""
    }

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#1e2a3f" : "transparent"
        border.color: "#1e2a3f"
        border.width: 1
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#9ca3af"
        font: parent.font

```

```

        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}
}
}

```

```

Rectangle {
    width: 300
    height: 500
    radius: 14
    color: "#0f1420"
    border.color: "#1e2a3f"
    border.width: 1

```

```

Column {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.margins: 20
    spacing: 4

```

```

Text {
    text: "KAYDEDİLEN NOKTALAR"
    color: "#6b7280"
    font.family: "Segoe UI"
    font.pixelSize: 11
    font.bold: true
    font.letterSpacing: 1
}

```

```

Text {
    text: "Düzeltilmek için bir noktaya tıkla"
    color: "#4b5563"
    font.family: "Segoe UI"
    font.pixelSize: 10
}

```

```

ListView {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.margins: 16
    anchors.topMargin: 60
    clip: true
    spacing: 8

    model: mesafeNoktalari

    delegate: Rectangle {
        width: ListView.view.width
        height: 48
        radius: 8
        color: mesafeSatirAlani.containsMouse ? "#241a38" : "#0a0e17"
        border.color: index === mesafeDuzenlemeIndex ? "#9333ea" : "#1e2a3f"
        border.width: index === mesafeDuzenlemeIndex ? 2 : 1
        Behavior on color { ColorAnimation { duration: 120 } }
    }
}

```

```

Row {
    anchors.fill: parent
    anchors.leftMargin: 12
    anchors.rightMargin: 12
    spacing: 8

    Rectangle {
        width: 22
        height: 22
        radius: 11
        anchors.verticalCenter: parent.verticalCenter
        color: "#241a38"
        border.color: "#9333ea"
        border.width: 1

        Text {
            anchors.centerIn: parent
            text: index + 1
            color: "#c084fc"
            font.pixelSize: 10
            font.bold: true
        }
    }
}

Column {
    anchors.verticalCenter: parent.verticalCenter
    spacing: 1

    Text {
        text: (modelData.hedefMm / 10).toFixed(1) + " cm"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }

    Text {
        text: "ADC: " + modelData.hamDeger.toFixed(0)
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 10
    }
}

Text {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.rightMargin: 12
    text: "↖"
    color: "#6b7280"
    font.pixelSize: 13
    visible: mesafeSatirAlani.containsMouse
}

MouseArea {
    id: mesafeSatirAlani
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
}

```

```

        onClicked: {
            hedefCmKutusu.text = (modelData.hedefMm / 10).toString()
            mesafeDuzenlemeIndex = index
        }
    }
}
}
}
}
}

// =====
// B) STROKE SINIRLARI – 2 nokta, sadelestirilmis
// =====
Item {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 30
    visible: mesafeAsama === 1

    Column {
        anchors.centerIn: parent
        spacing: 20
        width: 380
        visible: mesafeBAraEkranGoster

        Text {
            width: parent.width
            horizontalAlignment: Text.AlignHCenter
            wrapMode: Text.WordWrap
            text: "Bu sensör için kayıtlı bir stroke sınırı bulundu"
            color: "#dce8f5"
            font.family: "Segoe UI"
            font.pixelSize: 16
            font.bold: true
        }

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter
            width: 280
            height: 72
            radius: 12
            color: "#241a38"
            border.color: "#9333ea"
            border.width: 1

            Text {
                anchors.centerIn: parent
                text: "Üst: " + mesafeUstDeger.toFixed(1) + " mm    Alt: " +
mesafeAltDeger.toFixed(1) + " mm"
                color: "#c084fc"
                font.family: "Segoe UI"
                font.pixelSize: 15
                font.bold: true
            }
        }
    }

    Button {

```

```

        width: parent.width
        height: 44
        text: "Yeniden Kalibre Et"
        font.pixelSize: 14
        font.bold: true

        onClicked: {
            mesafeBAraEkranGoster = false
            mesafeAdimi = 0
        }

        background: Rectangle {
            radius: 8
            color: parent.hovered ? "#a855f7" : "#9333ea"
            Behavior on color { ColorAnimation { duration: 120 } }
        }

        contentItem: Text {
            text: parent.text
            color: "#ffffff"
            font: parent.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }
}

Column {
    anchors.centerIn: parent
    spacing: 20
    width: 360
    visible: !mesafeBAraEkranGoster

    Rectangle {
        anchors.horizontalCenter: parent.horizontalCenter
        visible: mesafeUstKayitliVar || mesafeAltKayitliVar
        width: sonKaliMetni3.implicitWidth + 20
        height: 26
        radius: 13
        color: "#0f1420"
        border.color: "#1e2a3f"
        border.width: 1

        Text {
            id: sonKaliMetni3
            anchors.centerIn: parent
            text: "Son kalibrasyon: " + mesafeSonTarih
            color: "#6b7280"
            font.pixelSize: 11
        }
    }

    Text {
        anchors.horizontalCenter: parent.horizontalCenter
        text: "Adım " + (mesafeAdimi + 1) + " / 2"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 11
        font.letterSpacing: 1
    }
}

```

```

Row {
    anchors.horizontalCenter: parent.horizontalCenter
    spacing: 6

    Repeater {
        model: 2
        Rectangle {
            width: index === mesafeAdimi ? 22 : 8
            height: 8
            radius: 4
            color: index <= mesafeAdimi ? "#9333ea" : "#1e2a3f"
            Behavior on width { NumberAnimation { duration: 150 } }
        }
    }
}

Column {
    width: parent.width
    spacing: 20
    visible: mesafeAdimi === 0

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Boruyu üst pozisyona getirin ve sabitleyin"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 14
    }

    Rectangle {
        anchors.horizontalCenter: parent.horizontalCenter
        width: 220
        height: 72
        radius: 12
        color: "#241a38"
        border.color: "#9333ea"
        border.width: 1

        Text {
            anchors.centerIn: parent
            text: sensorManager.konum.toFixed(1) + " mm"
            color: "#c084fc"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }
    }

    Button {
        width: parent.width
        height: 44
        text: "Üst Pozisyonu Kaydet"
        font.pixelSize: 14
        font.bold: true

        onClicked: {
            mesafeUstDeger = sensorManager.konum
        }
    }
}

```

```

        mesafeAdimi = 1
    }

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#a855f7" : "#9333ea"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Column {
    width: parent.width
    spacing: 16
    visible: mesafeAdimi == 1

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Boruyu alt pozisyona getirin ve sabitleyin"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 14
    }

    Rectangle {
        anchors.horizontalCenter: parent.horizontalCenter
        width: 220
        height: 72
        radius: 12
        color: "#241a38"
        border.color: "#9333ea"
        border.width: 1

        Text {
            anchors.centerIn: parent
            text: sensorManager.konum.toFixed(1) + " mm"
            color: "#c084fc"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }
    }
}

Button {
    width: parent.width
    height: 44
    text: "Alt Pozisyonu Kaydet ve Tamamla"
    font.pixelSize: 14
    font.bold: true

```

```

        onClicked: {
            mesafeAltDeger = sensorManager.konum
            if (Math.abs(mesafeUstDeger - mesafeAltDeger) < 30) {
                mesafeYakinUyariGoster = true
            } else {
                mesafeBKaydetVeBitir()
            }
        }
    }

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#22c55e" : "#16a34a"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Button {
    width: parent.width
    height: 38
    text: "← Geri"
    font.pixelSize: 13

    onClicked: mesafeAdimi = 0

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#1e2a3f" : "transparent"
        border.color: "#1e2a3f"
        border.width: 1
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#9ca3af"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

}

Rectangle {
    anchors.fill: parent
    color: "#000000b3"
    visible: mesafeYakinUyariGoster
    z: 200

    MouseArea { anchors.fill: parent }

    Rectangle {

```

```

anchors.centerIn: parent
width: 380
height: 200
radius: 14
color: "#0f1420"
border.color: "#1e2a3f"
border.width: 1

Column {
    anchors.centerIn: parent
    spacing: 18
    width: parent.width - 40

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Üst ve alt pozisyon birbirine çok yakın, emin misiniz?"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 14
        font.bold: true
    }

    Row {
        anchors.horizontalCenter: parent.horizontalCenter
        spacing: 12

        Button {
            width: 150
            height: 42
            text: "Vazgeç"

            onClicked: mesafeYakinUyariGoster = false

            background: Rectangle {
                radius: 8
                color: parent.hovered ? "#1e2a3f" : "transparent"
                border.color: "#1e2a3f"
                border.width: 1
                Behavior on color { ColorAnimation { duration: 120 } }
            }

            contentItem: Text {
                text: parent.text
                color: "#9ca3af"
                font: parent.font
                horizontalAlignment: Text.AlignHCenter
                verticalAlignment: Text.AlignVCenter
            }
        }

        Button {
            width: 150
            height: 42
            text: "Evet, Devam Et"
            font.bold: true

            onClicked: {
                mesafeYakinUyariGoster = false
            }
        }
    }
}

```

```
        mesafeBKaydetVeBitir()
    }

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#22c55e" : "#16a34a"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}
}
```

```

Rectangle {
    id: mesafeBildirimKutusu
    anchors.top: parent.top
    anchors.horizontalCenter: parent.horizontalCenter
    anchors.topMargin: 12
    z: 100
    width: mesafeBildirimMetni.implicitWidth + 24
    height: 32
    radius: 16
    color: mesafeBildirimBasarili ? "#123321" : "#3a1414"
    opacity: mesafeBildirimGoster ? 1 : 0
    visible: opacity > 0
    Behavior on opacity { NumberAnimation { duration: 200 } }
}

```

```
Text {
  id: mesafeBildirimMetni
  anchors.centerIn: parent
  text: mesafeBildirimMesaj
  color: mesafeBildirimBasarili ? "#4ade80" : "#f87171"
  font.family: "Segoe UI"
  font.pixelSize: 12
  font.bold: true
}
```

qml\pages\LoginPage.qml

345 satir

```
import QtQuick 6.7
import QtQuick.Controls 6.7

Rectangle {
    id: sayfa

    property bool sifreGorunur: false
    property bool beniHatirla: false

    signal girisBasarili()

    gradient: Gradient {
        GradientStop { position: 0.0; color: "#101c33" }
        GradientStop { position: 1.0; color: "#0a1220" }
    }

    Row {
        anchors.top: parent.top
        anchors.right: parent.right
        anchors.margins: 24
        spacing: 8

        Rectangle {
            width: 44
            height: 30
            radius: 6
            color: "#0f1420"
            border.color: "#3b82f6"
            border.width: 1

            Text {
                anchors.centerIn: parent
                text: "TR"
                color: "#ffffff"
                font.pixelSize: 12
                font.bold: true
            }
        }

        Rectangle {
            width: 44
            height: 30
            radius: 6
            color: "#0f1420"
            border.color: "#1e2a3f"
            border.width: 1

            Text {
                anchors.centerIn: parent
                text: "EN"
                color: "#6b7280"
                font.pixelSize: 12
            }
        }
    }
}
```

```
Rectangle {
    anchors.centerIn: kart
    width: kart.width + 48
    height: kart.height + 48
    radius: 28
    color: "#18000000"
}
```

```
Rectangle {
    anchors.centerIn: kart
    width: kart.width + 32
    height: kart.height + 32
    radius: 24
    color: "#25000000"
}
```

```
Rectangle {
    anchors.centerIn: kart
    width: kart.width + 18
    height: kart.height + 18
    radius: 20
    color: "#35000000"
}
```

```
Rectangle {
    anchors.centerIn: kart
    width: kart.width + 8
    height: kart.height + 8
    radius: 18
    color: "#45000000"
}
```

```
Rectangle {
    id: kart
    anchors.centerIn: parent
    width: 400
    color: "#0f1420"
    radius: 16
    border.color: "#4b5563"
    border.width: 1
    height: govde.implicitHeight + 48
}
```

```
Column {
    id: govde
    anchors.top: parent.top
    anchors.horizontalCenter: parent.horizontalCenter
    anchors.topMargin: 32
    width: parent.width - 48
    spacing: 20

    Image {
        anchors.horizontalCenter: parent.horizontalCenter
        source: "../assets/logo.png"
        width: 144
        height: 144
        fillMode: Image.PreserveAspectFit
    }

    Rectangle {
        width: parent.width
    }
}
```

```

        height: 1
        color: "#1e2a3f"
    }

    Column {
        width: parent.width
        spacing: 4

        Text {
            anchors.horizontalCenter: parent.horizontalCenter
            text: "Sisteme Giriş"
            color: "ffffff"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }

        Text {
            anchors.horizontalCenter: parent.horizontalCenter
            text: "Yetkili hesabınızla devam edin"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 12
        }
    }
}

Column {
    width: parent.width
    spacing: 6

    Text {
        text: "KULLANICI ADI"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 10
        font.letterSpacing: 1
    }

    TextField {
        id: kullaniciAdiKutusu
        width: parent.width
        height: 42
        placeholderText: "kullanici_adi"
        placeholderTextColor: "#4b5563"
        color: "ffffff"
        font.pixelSize: 13
        leftPadding: 12
        verticalAlignment: TextInput.AlignVCenter
        background: Rectangle {
            color: "#0a0e17"
            radius: 6
            border.color: kullaniciAdiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
            border.width: 1
        }
    }
}

Column {
    width: parent.width
    spacing: 6

```

```

Text {
    text: "ŞİFRE"
    color: "#6b7280"
    font.family: "Segoe UI"
    font.pixelSize: 10
    font.letterSpacing: 1
}

TextField {
    id: sifreKutusu
    width: parent.width
    height: 42
    placeholderText: "*****"
    placeholderTextColor: "#4b5563"
    echoMode: sayfa.sifreGorunur ? TextInput.Normal : TextInput.Password
    color: "#ffffff"
    font.pixelSize: 13
    leftPadding: 12
    rightPadding: 48
    verticalAlignment: TextInput.AlignVCenter
    background: Rectangle {
        color: "#0a0e17"
        radius: 6
        border.color: sifreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
        border.width: 1
    }
}

Rectangle {
    id: sifreGosterButonu
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.rightMargin: 8
    width: 28
    height: 28
    radius: 6
    color: sifreGosterMouse.pressed ? "#162033" : (sifreGosterMouse.containsMouse ?
"#111827" : "transparent")
    border.color: sayfa.sifreGorunur ? "#334155" : "transparent"
    border.width: 1

    property color ikonRengi: sifreGosterMouse.containsMouse || sayfa.sifreGorunur ?
"#cbd5e1" : "#6b7280"

    Canvas {
        id: sifreIkonu
        anchors.centerIn: parent
        width: 18
        height: 18

        onPaint: {
            var ctx = getContext("2d")
            ctx.clearRect(0, 0, width, height)
            ctx.lineWidth = 1.7
            ctx.lineCap = "round"
            ctx.lineJoin = "round"
            ctx.strokeStyle = sifreGosterButonu.ikonRengi

            ctx.beginPath()
            ctx.moveTo(1.5, 9)

```

```

        ctx.bezierCurveTo(4.2, 4.8, 7.0, 3.8, 9, 3.8)
        ctx.bezierCurveTo(11.0, 3.8, 13.8, 4.8, 16.5, 9)
        ctx.bezierCurveTo(13.8, 13.2, 11.0, 14.2, 9, 14.2)
        ctx.bezierCurveTo(7.0, 14.2, 4.2, 13.2, 1.5, 9)
        ctx.stroke()

        ctx.beginPath()
        ctx.arc(9, 9, 2.5, 0, Math.PI * 2, false)
        ctx.stroke()

        if (sayfa.sifreGorunur) {
            ctx.beginPath()
            ctx.moveTo(3, 15)
            ctx.lineTo(15, 3)
            ctx.stroke()
        }
    }

    Connections {
        target: sayfa
        function onSifreGorunurChanged() {
            sifreIkonu.requestPaint()
        }
    }

    Connections {
        target: sifreGosterButonu
        function onIkonRengiChanged() {
            sifreIkonu.requestPaint()
        }
    }
}

MouseArea {
    id: sifreGosterMouse
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
}

Text {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.rightMargin: 12
    text: sayfa.sifreGorunur ? "👁" : "👁"
    color: "#6b7280"
    font.pixelSize: 14
    visible: false

    MouseArea {
        anchors.fill: parent
        anchors.margins: -8
        cursorShape: Qt.PointingHandCursor
        onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
    }
}
}
}

```

```

Button {
    width: parent.width
    height: 44
    text: "Giriş Yap"
    font.family: "Segoe UI"
    font.pixelSize: 14
    font.bold: true

    onClicked: {
        var sonuc = backend.girisYap(kullaniciAdiKutusu.text, sifreKutusu.text)

        if (sonuc) {
            sayfa.girisBasarili()
        } else {
            console.log("Kullanici adi veya sifre hatali")
        }
    }

    background: Rectangle {
        radius: 8
        color: parent.pressed ? "#2563eb" : (parent.hovered ? "#4f8cf7" : "#3b82f6")
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Rectangle {
    width: parent.width
    height: 1
    color: "#1e2a3f"
}

Text {
    anchors.horizontalCenter: parent.horizontalCenter
    text: "\u00A9 2026 Liya Test A.Ş. SLIPER Analiz Yazılımı"
    color: "#4b5563"
    font.family: "Segoe UI"
    font.pixelSize: 10
}
}
}
}

```

1144 satır

```
import QtQuick 6.7
import QtQuick.Controls 6.7
import QtCharts 6.7

Rectangle {
    color: "#0a0e17"
    property real zamanSayaci: 0
    property int aktifOlcumId: -1
    property string uyariMesaji: ""

    property var basincGecmis: []
    property var konumGecmis: []
    property var hizGecmis: []
    property var debiGecmis: []

    signal olcumTamamlandi(int id)

    function eksenTavaniHesapla(dizi, tavan, taban) {
        if (dizi.length === 0) return taban
        var maxDeger = Math.max.apply(null, dizi)
        var hedef = maxDeger * 1.25
        if (hedef < taban) hedef = taban
        if (hedef > tavan) hedef = tavan
        return hedef
    }

    // Basınc (mbar) atmosferik seviyede sabit bir taban etrafında dalgalanır,
    // bu yüzden sabit tavanlı ölçekleme yerine veriye göre kayan min/max kullanılır.
    function basincEkseniHesapla(dizi) {
        if (dizi.length === 0) return { min: 0, max: 1100 }
        var maxDeger = Math.max.apply(null, dizi)
        var minDeger = Math.min.apply(null, dizi)
        var pad = Math.max((maxDeger - minDeger) * 0.2, maxDeger * 0.02, 5)
        return { min: Math.max(0, minDeger - pad), max: maxDeger + pad }
    }

    Connections {
        target: sensorManager
        function onVeriGuncellendi() {
            if (!sensorManager.veriGecerli || aktifOlcumId <= 0 || calculator.duraklatildi) return

            zamanSayaci += 0.2

            basincSerisi.append(zamanSayaci, sensorManager.basinc)
            konumSerisi.append(zamanSayaci, sensorManager.konum)
            hizSerisi.append(zamanSayaci, sensorManager.hiz)
            debiSerisi.append(zamanSayaci, sensorManager.debi)

            basincGecmis.push(sensorManager.basinc)
            konumGecmis.push(sensorManager.konum)
            hizGecmis.push(sensorManager.hiz)
            debiGecmis.push(sensorManager.debi)

            if (zamanSayaci > 60) {
                basincSerisi.remove(0)
                konumSerisi.remove(0)
            }
        }
    }
}
```

```

        hizSerisi.remove(0)
        debiSerisi.remove(0)

        basincGecmis.shift()
        konumGecmis.shift()
        hizGecmis.shift()
        debiGecmis.shift()

        xEkseni.min = zamanSayaci - 60
        xEkseni.max = zamanSayaci
        konumXEkseni.min = zamanSayaci - 60
        konumXEkseni.max = zamanSayaci
        hizXEkseni.min = zamanSayaci - 60
        hizXEkseni.max = zamanSayaci
        debiXEkseni.min = zamanSayaci - 60
        debiXEkseni.max = zamanSayaci
    }

    var basincAralik = basincEkseniHesapla(basincGecmis)
    yEkseni.min = basincAralik.min
    yEkseni.max = basincAralik.max
    konumYEkseni.max = eksenTavaniHesapla(konumGecmis, 500, 50)
    hizYEkseni.max = eksenTavaniHesapla(hizGecmis, 4, 0.5)
    debiYEkseni.max = eksenTavaniHesapla(debiGecmis, 180, 10)

    calculator.konumGuncelle(sensorManager.konum)
}

Row {
    anchors.fill: parent
    spacing: 0

    Rectangle {
        id: testBilgileriPaneli
        width: 280
        height: parent.height
        color: "#0f1420"

        Text {
            anchors.top: parent.top
            anchors.left: parent.left
            anchors.margins: 20
            text: "TEST BİLGİLERİ"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 12
            font.bold: true
            font.letterSpacing: 1
        }

        Column {
            id: formAlanlari
            anchors.top: parent.top
            anchors.left: parent.left
            anchors.right: parent.right
            anchors.margins: 20
            anchors.topMargin: 60
            spacing: 6

```

```

Text {
    text: "Müşteri"
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 12
}

TextField {
    id: musterikutusu
    width: parent.width
    height: 38
    placeholderText: "Musteri adi girin..."
    placeholderTextColor: "#4b5563"
    color: "#ffffff"
    font.pixelSize: 13
    leftPadding: 10
    verticalAlignment: TextInput.AlignVCenter
    background: Rectangle {
        color: "#0a0e17"
        radius: 6
        border.color: musterikutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
        border.width: 1
    }
}

Text {
    text: "Beton Reçetesi"
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 12
}

ComboBox {
    id: receteKutusu
    width: parent.width
    height: 38
    model: ["Reçete Seçin...", "C25/30 Standart", "C30/37 SCC", "C35/45 Yüksek Mukavemet"]

    background: Rectangle {
        color: "#0a0e17"
        radius: 6
        border.color: receteKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
        border.width: 1
    }

    contentItem: Text {
        text: receteKutusu.displayText
        color: "#9ca3af"
        font.pixelSize: 13
        leftPadding: 10
        verticalAlignment: Text.AlignVCenter
    }
}

Text {
    text: "Eklenen Ağırlık (kg)"
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 12
}

```

```

TextField {
    id: agirlikKutusu
    width: parent.width
    height: 38
    placeholderText: "örn: 4.8"
    placeholderTextColor: "#4b5563"
    color: "ffffff"
    font.pixelSize: 13
    leftPadding: 10
    verticalAlignment: TextInput.AlignVCenter
    validator: DoubleValidator { bottom: 0; top: 20; decimals: 1 }
    background: Rectangle {
        color: "#0a0e17"
        radius: 6
        border.color: agirlikKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
        border.width: 1
    }
}

```

```

Item { width: parent.width; height: 10 }

```

```

Button {
    id: baslatButonu
    width: parent.width
    height: 44
    text: "► Ölçümü Başlat"
    font.family: "Segoe UI"
    font.pixelSize: 14
    font.bold: true
    enabled: aktifOlcumId <= 0

    onClicked: {
        if (!sensorManager.veriGecerli) {
            console.warn("Gerçek sensor verisi yok, ölçüm baslatılmadı.")
            uyariMesaji = "Cihaz bağlı değil. Lütfen önce sol alttan SLIPER-ESP32'ye
bağlanın."

            return
        }

        uyariMesaji = ""
        zamanSayaci = 0
        basincSerisi.clear()
        konumSerisi.clear()
        hizSerisi.clear()
        debiSerisi.clear()
        basincGecmis = []
        konumGecmis = []
        hizGecmis = []
        debiGecmis = []
        xEkseni.min = 0
        xEkseni.max = 60
        yEkseni.min = 0
        yEkseni.max = 1100
        konumXEkseni.min = 0
        konumYEkseni.max = 60
        konumYEkseni.max = 500
        hizXEkseni.min = 0
        hizXEkseni.max = 60
        hizYEkseni.max = 4
    }
}

```

```

        debiXEkseni.min = 0
        debiXEkseni.max = 60
        debiYEkseni.max = 180

        calculator.sifirla()
        aktifOlcumId = database.olcumBaslat(
            musterikutusu.text,
            receteKutusu.currentText,
            parseFloat(agirlikKutusu.text)
        )
        console.log("Aktif olcum id:", aktifOlcumId)
    }

    background: Rectangle {
        radius: 8
        color: !baslatButonu.enabled ? "#14532d" : (baslatButonu.pressed ? "#15803d" :
(baslatButonu.hovered ? "#22c55e" : "#16a34a"))
        opacity: baslatButonu.enabled ? 1.0 : 0.5
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: baslatButonu.text
        color: "#ffffff"
        opacity: baslatButonu.enabled ? 1.0 : 0.6
        font: baslatButonu.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Text {
    width: parent.width
    visible: uyariMesaji.length > 0
    text: uyariMesaji
    color: "#f87171"
    font.family: "Segoe UI"
    font.pixelSize: 11
    wrapMode: Text.WordWrap
}

Row {
    width: parent.width
    spacing: 10

    Button {
        id: duraklatButonu
        width: (parent.width - parent.spacing) / 2
        height: 40
        text: calculator.duraklatildi ? "► Devam Et" : "|| Duraklat"
        font.family: "Segoe UI"
        font.pixelSize: 13

        onClicked: {
            if (calculator.duraklatildi) {
                calculator.devamEt()
            } else {
                calculator.duraklat()
            }
        }
    }
}

```

```

"#92400e")

background: Rectangle {
    radius: 8
    color: duraklatButonu.pressed ? "#78350f" : (duraklatButonu.hovered ? "#a16207" :

background: Rectangle {
    radius: 8
    color: bitirButonu.pressed ? "#7f1d1d" : (bitirButonu.hovered ? "#b91c1c" :

background: Rectangle {
    radius: 8
    color: duraklatButonu.pressed ? "#78350f" : (duraklatButonu.hovered ? "#a16207" :
    border.color: "#d97706"
    border.width: 1
}

contentItem: Text {
    text: duraklatButonu.text
    color: "#ffffff"
    font: duraklatButonu.font
    horizontalAlignment: Text.AlignHCenter
    verticalAlignment: Text.AlignVCenter
}
}

Button {
    id: bitirButonu
    width: (parent.width - parent.spacing) / 2
    height: 40
    text: "■ Bitir"
    font.family: "Segoe UI"
    font.pixelSize: 13

    onClicked: {
        if (aktifOlcumId > 0) {
            bitirOnayPopup.open()
        } else {
            uyariMesaji = "Aktif bir ölçüm yok. Önce ölçümü başlatın."
        }
    }

background: Rectangle {
    radius: 8
    color: bitirButonu.pressed ? "#7f1d1d" : (bitirButonu.hovered ? "#b91c1c" :

background: Rectangle {
    radius: 8
    color: bitirButonu.pressed ? "#7f1d1d" : (bitirButonu.hovered ? "#b91c1c" :
    border.color: "#dc2626"
    border.width: 1
}

contentItem: Text {
    text: bitirButonu.text
    color: "#ffffff"
    font: bitirButonu.font
    horizontalAlignment: Text.AlignHCenter
    verticalAlignment: Text.AlignVCenter
}
}
}

Rectangle {
    id: durumKarti
    anchors.top: formAlanlari.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.topMargin: 20
    anchors.leftMargin: 20

```

```

anchors.rightMargin: 20
height: 52
radius: 10
color: "#0a0e17"
border.color: "#1e2a3f"
border.width: 1

Row {
    anchors.left: parent.left
    anchors.verticalCenter: parent.verticalCenter
    anchors.leftMargin: 14
    spacing: 10

    Rectangle {
        id: trafikIsigi
        width: 12
        height: 12
        radius: 6
        anchors.verticalCenter: parent.verticalCenter
        color: {
            if (!sensorManager.veriGecerli) return "#dc2626"
            if (calculator.duraklatildi) return "#6b7280"
            if (calculator.durum == "YUKARIDA") return "#16a34a"
            if (calculator.durum == "INIYOR") return "#f59e0b"
            return "#dc2626"
        }

        Rectangle {
            anchors.centerIn: parent
            width: parent.width + 8
            height: parent.height + 8
            radius: width / 2
            color: "transparent"
            border.color: parent.color
            border.width: 1
            opacity: 0.35
        }
    }

    Text {
        anchors.verticalCenter: parent.verticalCenter
        text: {
            if (!sensorManager.veriGecerli) return "VERI YOK"
            if (calculator.duraklatildi) return "DURAKLATILDI"
            return calculator.durum
        }
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }
}

Row {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.rightMargin: 14
    spacing: 6

    Text {

```

```

        anchors.verticalCenter: parent.verticalCenter
        text: "STROKE"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 10
        font.letterSpacing: 1
    }

    Text {
        anchors.verticalCenter: parent.verticalCenter
        text: calculator.strokeSayisi
        color: "#e8a020"
        font.family: "Segoe UI"
        font.pixelSize: 16
        font.bold: true
    }
}

Text {
    id: anlikDegerlerBaslik
    anchors.top: durumKarti.bottom
    anchors.left: parent.left
    anchors.topMargin: 24
    anchors.leftMargin: 20
    text: "ANLIK DEĞERLER"
    color: "#6b7280"
    font.family: "Segoe UI"
    font.pixelSize: 12
    font.bold: true
    font.letterSpacing: 1
}

Grid {
    anchors.top: anlikDegerlerBaslik.bottom
    anchors.left: parent.left
    anchors.topMargin: 12
    anchors.leftMargin: 20
    columns: 2
    spacing: 10

    Rectangle {
        id: basincKarti
        width: 115
        height: 90
        radius: 10
        color: "#0a0e17"
        border.color: "#1e2a3f"
        border.width: 1
        clip: true

        Rectangle { width: 4; height: parent.height; color: "#3b82f6" }

        Column {
            anchors.top: parent.top
            anchors.right: parent.right
            anchors.bottom: parent.bottom
            anchors.left: parent.left
            anchors.topMargin: 10
            anchors.rightMargin: 10

```

```

        anchors.bottomMargin: 10
        anchors.leftMargin: 16
        spacing: 6

        Text {
            text: "BASINÇ"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            text: sensorManager.veriGecerli ? sensorManager.basinc.toFixed(1) : "--"
            color: "#3b82f6"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }

        Text {
            text: "mbar"
            color: "#4b5563"
            font.family: "Segoe UI"
            font.pixelSize: 10
        }
    }
}

Rectangle {
    width: 115
    height: 90
    radius: 10
    color: "#0a0e17"
    border.color: "#1e2a3f"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#9333ea" }

    Column {
        anchors.top: parent.top
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.left: parent.left
        anchors.topMargin: 10
        anchors.rightMargin: 10
        anchors.bottomMargin: 10
        anchors.leftMargin: 16
        spacing: 6

        Text {
            text: "KONUM"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {

```

```

        text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) : "--"
        color: "#9333ea"
        font.family: "Segoe UI"
        font.pixelSize: 22
        font.bold: true
    }

    Text {
        text: "mm"
        color: "#4b5563"
        font.family: "Segoe UI"
        font.pixelSize: 10
    }
}

}

Rectangle {
    width: 115
    height: 90
    radius: 10
    color: "#0a0e17"
    border.color: "#1e2a3f"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#f59e0b" }

    Column {
        anchors.top: parent.top
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.left: parent.left
        anchors.topMargin: 10
        anchors.rightMargin: 10
        anchors.bottomMargin: 10
        anchors.leftMargin: 16
        spacing: 6

        Text {
            text: "HIZ"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) : "--"
            color: "#f59e0b"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }

        Text {
            text: "m/s"
            color: "#4b5563"
            font.family: "Segoe UI"
            font.pixelSize: 10
        }
    }
}

```

```

    }
}

Rectangle {
    width: 115
    height: 90
    radius: 10
    color: "#0a0e17"
    border.color: "#1e2a3f"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#16a34a" }

    Column {
        anchors.top: parent.top
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.left: parent.left
        anchors.topMargin: 10
        anchors.rightMargin: 10
        anchors.bottomMargin: 10
        anchors.leftMargin: 16
        spacing: 6

        Text {
            text: "DEBİ"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) : "--"
            color: "#16a34a"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }

        Text {
            text: "m3/h"
            color: "#4b5563"
            font.family: "Segoe UI"
            font.pixelSize: 10
        }
    }
}

}

Rectangle {
    id: panelAyraci
    width: 3
    height: parent.height
    color: "#05070b"

    Rectangle {
        anchors.right: parent.right

```

```

        anchors.top: parent.top
        anchors.bottom: parent.bottom
        width: 1
        color: "#403b82f6"
    }
}

Item {
    width: parent.width - testBilgileriPaneli.width - panelAyraci.width
    height: parent.height

    Grid {
        anchors.fill: parent
        anchors.margins: 16
        columns: 2
        spacing: 16

        Rectangle {
            width: (parent.width - parent.spacing) / 2
            height: (parent.height - parent.spacing) / 2
            radius: 10
            color: "#0f1420"
            border.color: "#17263d"
            border.width: 1

            Item {
                id: basincBaslik
                anchors.top: parent.top
                anchors.left: parent.left
                anchors.right: parent.right
                anchors.margins: 14
                height: 20

                Rectangle {
                    id: basincNoktasi
                    width: 8
                    height: 8
                    radius: 4
                    color: "#3b82f6"
                    anchors.left: parent.left
                    anchors.verticalCenter: parent.verticalCenter
                }

                Text {
                    anchors.left: basincNoktasi.right
                    anchors.leftMargin: 8
                    anchors.verticalCenter: parent.verticalCenter
                    text: "Basınç - Zaman"
                    color: "#9ca3af"
                    font.family: "Segoe UI"
                    font.pixelSize: 13
                    font.bold: true
                }

                Text {
                    anchors.right: parent.right
                    anchors.verticalCenter: parent.verticalCenter
                    text: sensorManager.veriGecerli ? sensorManager.basinc.toFixed(1) + " mbar" : "--
mbar"

                    color: "#3b82f6"

```

```

        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }
}

ChartView {
    anchors.top: basincBaslik.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 4
    anchors.bottomMargin: 4
    backgroundColor: "transparent"
    legend.visible: false
    antialiasing: true

    ValueAxis {
        id: xEkseni
        min: 0
        max: 60
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    ValueAxis {
        id: yEkseni
        min: 0
        max: 1100
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    LineSeries {
        id: basincSerisi
        axisX: xEkseni
        axisY: yEkseni
        color: "#3b82f6"
        width: 2
    }
}

Rectangle {
    width: (parent.width - parent.spacing) / 2
    height: (parent.height - parent.spacing) / 2
    radius: 10
    color: "#0f1420"
    border.color: "#241a38"
    border.width: 1

    Item {
        id: konumBaslik
        anchors.top: parent.top
    }
}

```

```
anchors.left: parent.left
anchors.right: parent.right
anchors.margins: 14
height: 20
```

```
Rectangle {
    id: konumNoktasi
    width: 8
    height: 8
    radius: 4
    color: "#9333ea"
    anchors.left: parent.left
    anchors.verticalCenter: parent.verticalCenter
}
```

```
Text {
    anchors.left: konumNoktasi.right
    anchors.leftMargin: 8
    anchors.verticalCenter: parent.verticalCenter
    text: "Konum - Zaman"
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}
```

```
Text {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) + " mm" : "--

    color: "#9333ea"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}
```

```
}

mm"

ChartView {
    anchors.top: konumBaslik.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 4
    anchors.bottomMargin: 4
    backgroundColor: "transparent"
    legend.visible: false
    antialiasing: true

    ValueAxis {
        id: konumXEkseni
        min: 0
        max: 60
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }
}
```

```

ValueAxis {
    id: konumYEkseni
    min: 0
    max: 500
    gridLineColor: "#182131"
    labelsColor: "#4b5563"
    labelsFont.pixelSize: 9
    lineVisible: false
    minorGridVisible: false
}

LineSeries {
    id: konumSerisi
    axisX: konumXEkseni
    axisY: konumYEkseni
    color: "#9333ea"
    width: 2
}

Connections {
    target: calculator
    function onStrokeSayisiChanged() {
        if (aktifOlcumId > 0 && sensorManager.veriGecerli) {
            database.strokeKaydet(aktifOlcumId, sensorManager.basinc,
sensorManager.konum, sensorManager.debi, true)
        }
    }
}
}
}

```

```

Rectangle {
    width: (parent.width - parent.spacing) / 2
    height: (parent.height - parent.spacing) / 2
    radius: 10
    color: "#0f1420"
    border.color: "#3a2a14"
    border.width: 1
}

```

```

Item {
    id: hizBaslik
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.margins: 14
    height: 20
}

```

```

Rectangle {
    id: hizNoktasi
    width: 8
    height: 8
    radius: 4
    color: "#f59e0b"
    anchors.left: parent.left
    anchors.verticalCenter: parent.verticalCenter
}

```

```

Text {
    anchors.left: hizNoktasi.right
    anchors.leftMargin: 8
}

```

```

        anchors.verticalCenter: parent.verticalCenter
        text: "Hız - Zaman"
        color: "#9ca3af"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }

```

```

Text {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) + " m/s" : "--

    color: "#f59e0b"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}

```

m/s"

```

ChartView {
    anchors.top: hizBaslik.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 4
    anchors.bottomMargin: 4
    backgroundColor: "transparent"
    legend.visible: false
    antialiasing: true

    ValueAxis {
        id: hizXEkseni
        min: 0
        max: 60
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    ValueAxis {
        id: hizYEkseni
        min: 0
        max: 4
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    LineSeries {
        id: hizSerisi
        axisX: hizXEkseni
        axisY: hizYEkseni
        color: "#f59e0b"
        width: 2
    }
}

```

```

    }
}

Rectangle {
    width: (parent.width - parent.spacing) / 2
    height: (parent.height - parent.spacing) / 2
    radius: 10
    color: "#0f1420"
    border.color: "#163321"
    border.width: 1

    Item {
        id: debiBaslik
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.margins: 14
        height: 20

        Rectangle {
            id: debiNoktasi
            width: 8
            height: 8
            radius: 4
            color: "#16a34a"
            anchors.left: parent.left
            anchors.verticalCenter: parent.verticalCenter
        }

        Text {
            anchors.left: debiNoktasi.right
            anchors.leftMargin: 8
            anchors.verticalCenter: parent.verticalCenter
            text: "Debi - Zaman"
            color: "#9ca3af"
            font.family: "Segoe UI"
            font.pixelSize: 13
            font.bold: true
        }

        Text {
            anchors.right: parent.right
            anchors.verticalCenter: parent.verticalCenter
            text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) + " m3/h" : "--

m3/h"

            color: "#16a34a"
            font.family: "Segoe UI"
            font.pixelSize: 13
            font.bold: true
        }
    }
}

ChartView {
    anchors.top: debiBaslik.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 4
    anchors.bottomMargin: 4
    backgroundColor: "transparent"

```

```
legend.visible: false
antialiasing: true
```

```
ValueAxis {
    id: debiXEkseni
    min: 0
    max: 60
    gridLineColor: "#182131"
    labelsColor: "#4b5563"
    labelsFont.pixelSize: 9
    lineVisible: false
    minorGridVisible: false
}
```

```
ValueAxis {
    id: debiYEkseni
    min: 0
    max: 180
    gridLineColor: "#182131"
    labelsColor: "#4b5563"
    labelsFont.pixelSize: 9
    lineVisible: false
    minorGridVisible: false
}
```

```
LineSeries {
    id: debiSerisi
    axisX: debiXEkseni
    axisY: debiYEkseni
    color: "#16a34a"
    width: 2
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
Popup {
    id: bitirOnayPopup
    modal: true
    focus: true
    anchors.centerIn: parent
    width: 340
    padding: 24
    closePolicy: Popup.CloseOnEscape | Popup.CloseOnPressOutside

    background: Rectangle {
        color: "#0f1420"
        radius: 12
        border.color: "#1e2a3f"
        border.width: 1
    }

    Overlay.modal: Rectangle {
        color: "#a6000000"
    }

    contentItem: Column {
        spacing: 20
    }
}
```

```

Text {
    width: 292
    text: "Ölçümü bitirmek istediğinize emin misiniz?"
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 14
    font.bold: true
    wrapMode: Text.WordWrap
}

Row {
    width: 292
    spacing: 10

    Button {
        id: vazgecButonu
        width: (parent.width - parent.spacing) / 2
        height: 40
        text: "Vazgeç"
        font.family: "Segoe UI"
        font.pixelSize: 13

        onClicked: bitirOnayPopup.close()

        background: Rectangle {
            radius: 8
            color: vazgecButonu.pressed ? "#1e2a3f" : (vazgecButonu.hovered ? "#243349" :
"#182131")

            border.color: "#2c3b52"
            border.width: 1
        }

        contentItem: Text {
            text: vazgecButonu.text
            color: "#dce8f5"
            font: vazgecButonu.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }

    Button {
        id: evetBitirButonu
        width: (parent.width - parent.spacing) / 2
        height: 40
        text: "Evet, Bitir"
        font.family: "Segoe UI"
        font.pixelSize: 13

        onClicked: {
            bitirOnayPopup.close()
            var bitenId = aktifOlcumId
            aktifOlcumId = -1
            uyariMesaji = ""
            console.log("Olcum sonlandırıldı, id:", bitenId)
            olcumTamamlandi(bitenId)
        }

        background: Rectangle {

```

```
radius: 8
color: evetBitirButonu.pressed ? "#7f1d1d" : (evetBitirButonu.hovered ? "#b91c1c" :
"#991b1b")

border.color: "#dc2626"
border.width: 1
}

contentItem: Text {
    text: evetBitirButonu.text
    color: "#ffffff"
    font: evetBitirButonu.font
    horizontalAlignment: Text.AlignHCenter
    verticalAlignment: Text.AlignVCenter
}
}
}
}
```

862 satir

```

import QtQuick 6.7
import QtQuick.Controls 6.7
import QtCharts 6.7

Rectangle {
    color: "#0a0e17"

    property int olcumId: -1
    property real hesaplananTau0: 0
    property real hesaplananMu: 0
    property int playbackAdimi: -1
    property bool oynatiliyor: false
    property int playbackHizi: 1
    property string musteriAdi: ""
    property string receteAdi: ""

    onOlcumIdChanged: verileriYukle()

    function verileriYukle() {
        if (olcumId <= 0) {
            return
        }

        playbackAdimi = -1
        oynatiliyor = false
        playbackTimer.stop()

        var bilgi = database.olcumBilgisiGetir(olcumId)
        musteriAdi = bilgi.bulundu ? bilgi.musteri : "Bilinmiyor"
        receteAdi = bilgi.bulundu ? bilgi.recete : "Bilinmiyor"

        var sonuc = database.binghamHesapla(olcumId)

        hesaplananTau0 = sonuc.tau0
        hesaplananMu = sonuc.mu

        tau0Metni.text = sonuc.tau0.toFixed(2) + " mbar"
        muMetni.text = sonuc.mu.toFixed(2) + " mbar·h/m³"
        r2Metni.text = sonuc.r2.toFixed(2)

        durumKutusu.durumIyiMi = sonuc.yeterliVeri && sonuc.tau0 < 5 && sonuc.mu < 10
        durumKutusu.tau0Yuksek = sonuc.tau0 >= 5
        durumKutusu.muYuksek = sonuc.mu >= 10

        pqSerisi.clear()
        regresyonCizgisi.clear()
        oynatmaSerisi.clear()
        strokeModeli.clear()

        var strokeListesi = database.strokeVerileriGetir(olcumId)
        for (var i = 0; i < strokeListesi.length; i++) {
            var s = strokeListesi[i]
            pqSerisi.append(s.debi, s.basinc)
            strokeModeli.append({ stroke: s.stroke, p: s.basinc, q: s.debi, gecerli: s.gecerli })
        }
    }
}

```

```

        regresyonCizgisi.append(0, sonuc.tau0)
        regresyonCizgisi.append(20, sonuc.tau0 + sonuc.mu * 20)
    }

    Timer {
        id: playbackTimer
        interval: 1000 / playbackHizi
        repeat: true
        running: false

        onTriggered: {
            if (playbackAdimi < strokeModeli.count - 1) {
                playbackAdimi++
                var satir = strokeModeli.get(playbackAdimi)
                oynatmaSerisi.clear()
                oynatmaSerisi.append(satir.q, satir.p)
            } else {
                oynatiliyor = false
                playbackTimer.stop()
            }
        }
    }
}

Row {
    anchors.fill: parent
    spacing: 0

    Rectangle {
        id: ozetPaneli
        width: 280
        height: parent.height
        color: "#0f1420"

        Text {
            anchors.top: parent.top
            anchors.left: parent.left
            anchors.margins: 20
            text: "SONUÇ ÖZETİ"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 12
            font.bold: true
            font.letterSpacing: 1
        }

        Flickable {
            anchors.top: parent.top
            anchors.left: parent.left
            anchors.right: parent.right
            anchors.bottom: parent.bottom
            anchors.margins: 20
            anchors.topMargin: 60
            contentWidth: width
            contentHeight: sonucKartlari.height
            clip: true

            Column {
                id: sonucKartlari
                width: parent.width
                spacing: 10
            }
        }
    }
}

```

```

Rectangle {
    width: parent.width
    height: 70
    radius: 10
    color: "#0a0e17"
    border.color: "#17263d"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#3b82f6" }

    Column {
        anchors.left: parent.left
        anchors.verticalCenter: parent.verticalCenter
        anchors.leftMargin: 20
        spacing: 2

        Text {
            text: "AKMA GERİLMESİ ( $\tau_0$ )"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            id: tau0Metni
            text: "—"
            color: "#3b82f6"
            font.family: "Segoe UI"
            font.pixelSize: 20
            font.bold: true
        }
    }
}

Rectangle {
    width: parent.width
    height: 70
    radius: 10
    color: "#0a0e17"
    border.color: "#241a38"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#9333ea" }

    Column {
        anchors.left: parent.left
        anchors.verticalCenter: parent.verticalCenter
        anchors.leftMargin: 20
        spacing: 2

        Text {
            text: "PLASTİK VİSKOZİTE ( $\mu$ )"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }
    }
}

```

```

    }

    Text {
        id: muMetni
        text: "—"
        color: "#9333ea"
        font.family: "Segoe UI"
        font.pixelSize: 20
        font.bold: true
    }
}

Rectangle {
    width: parent.width
    height: 70
    radius: 10
    color: "#0a0e17"
    border.color: "#163321"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#16a34a" }

    Column {
        anchors.left: parent.left
        anchors.verticalCenter: parent.verticalCenter
        anchors.leftMargin: 20
        spacing: 2

        Text {
            text: "UYUM KALİTESİ ( $R^2$ )"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            id: r2Metni
            text: "—"
            color: "#16a34a"
            font.family: "Segoe UI"
            font.pixelSize: 20
            font.bold: true
        }
    }
}

Rectangle {
    width: parent.width
    radius: 10
    color: "#0a0e17"
    border.color: "#1e2a3f"
    border.width: 1
    height: tahminIcerik.implicitHeight + 24

    Column {
        id: tahminIcerik
        anchors.top: parent.top
    }
}

```

```
anchors.left: parent.left
anchors.right: parent.right
anchors.margins: 12
spacing: 8
```

```
Text {
    text: "BORU HATTI TAHMİNİ"
    color: "#6b7280"
    font.family: "Segoe UI"
    font.pixelSize: 10
    font.letterSpacing: 1
}
```

```
TextField {
    id: uzunlukKutusu
    width: parent.width
    height: 34
    placeholderText: "Boru Uzunluğu (m)"
    placeholderTextColor: "#4b5563"
    color: "#ffffff"
    font.pixelSize: 12
    leftPadding: 10
    verticalAlignment: TextInput.AlignVCenter
    validator: DoubleValidator { bottom: 0; top: 2000; decimals: 0 }
    background: Rectangle {
        color: "#060d17"
        radius: 6
        border.color: uzunlukKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
        border.width: 1
    }
}
```

```
TextField {
    id: debiKutusu
    width: parent.width
    height: 34
    placeholderText: "Hedef Debi (m³/h)"
    placeholderTextColor: "#4b5563"
    color: "#ffffff"
    font.pixelSize: 12
    leftPadding: 10
    verticalAlignment: TextInput.AlignVCenter
    validator: DoubleValidator { bottom: 0; top: 200; decimals: 1 }
    background: Rectangle {
        color: "#060d17"
        radius: 6
        border.color: debiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
        border.width: 1
    }
}
```

```
Button {
    width: parent.width
    height: 36
    text: "Tahmini Hesapla"
    font.pixelSize: 12
    font.bold: true

    background: Rectangle {
        radius: 6
    }
}
```

```

        color: parent.hovered ? "#4f8cf7" : "#3b82f6"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }

    onClicked: {
        var L = parseFloat(uzunlukKutusu.text)
        var Q = parseFloat(debiKutusu.text)

        if (isNaN(L) || isNaN(Q) || L <= 0) {
            tahminSonucu.text = "Geçerli değer girin"
            return
        }

        var basincMbar = (hesaplananTau0 + hesaplananMu * Q) * (L / 50)
        var basincBar = basincMbar / 1000
        tahminSonucu.text = basincBar.toFixed(2) + " bar"
    }
}

Text {
    id: tahminSonucu
    width: parent.width
    text: "-"
    color: "#e8a020"
    font.family: "Segoe UI"
    font.pixelSize: 18
    font.bold: true
    horizontalAlignment: Text.AlignHCenter
}
}

Rectangle {
    id: durumKutusu
    width: parent.width
    height: durumIcerik.implicitHeight + 24
    radius: 10
    color: durumIyiMi ? "#0f2417" : "#2a1414"
    border.color: durumIyiMi ? "#16a34a" : "#dc2626"
    border.width: 1

    property bool durumIyiMi: true
    property bool tau0Yuksek: false
    property bool muYuksek: false

    Column {
        id: durumIcerik
        anchors.top: parent.top
        anchors.horizontalCenter: parent.horizontalCenter
        anchors.topMargin: 12
        spacing: 8
        width: parent.width - 24
    }
}

```

```

Row {
    anchors.horizontalCenter: parent.horizontalCenter
    spacing: 10

    Rectangle {
        width: 24
        height: 24
        radius: 12
        anchors.verticalCenter: parent.verticalCenter
        color: durumKutusu.durumIyiMi ? "#16a34a" : "#dc2626"

        Text {
            anchors.centerIn: parent
            text: durumKutusu.durumIyiMi ? "✓" : "X"
            color: "#ffffff"
            font.pixelSize: 13
            font.bold: true
        }
    }

    Text {
        anchors.verticalCenter: parent.verticalCenter
        text: durumKutusu.durumIyiMi ? "POMPALANABİLİR" : "POMPALAMA SORUNU"
        color: durumKutusu.durumIyiMi ? "#4ade80" : "#f87171"
        font.family: "Segoe UI"
        font.pixelSize: 15
        font.bold: true
    }
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    visible: !durumKutusu.durumIyiMi
    text: {
        if (durumKutusu.tau0Yuksek && durumKutusu.muYuksek) {
            return "Hem akma gerilmesi hem plastik viskozite yüksek. Su/çimento oranını artırın ve agrega gradasyonunu gözden geçirin."
        } else if (durumKutusu.tau0Yuksek) {
            return "Akma gerilmesi yüksek. Su/çimento oranını artırmayı veya süperakışkanlaştırıcı dozajını yükseltmeyi değerlendirin."
        } else if (durumKutusu.muYuksek) {
            return "Plastik viskozite yüksek. İnce agrega oranını azaltmayı veya su azaltıcı katkı eklemeyi değerlendirin."
        }
        return "Yeterli stroke verisi bulunamadı."
    }
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 11
    wrapMode: Text.WordWrap
}

Button {
    width: parent.width
    height: 40
    text: "📄 PDF Rapor Oluştur"

```

```

font.pixelSize: 13
font.bold: true
enabled: olcumId > 0

onClicked: {
    reportManager.pdfOlustur(
        olcumId,
        musteriAdi,
        receteAdi,
        hesaplananTau0,
        hesaplananMu,
        parseFloat(r2Metni.text)
    )
    console.log("PDF rapor olusturuldu.")
}

background: Rectangle {
    radius: 8
    color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") : "#1e2a3f"
    Behavior on color { ColorAnimation { duration: 120 } }
}

contentItem: Text {
    text: parent.text
    color: "#ffffff"
    font: parent.font
    horizontalAlignment: Text.AlignHCenter
    verticalAlignment: Text.AlignVCenter
}
}

Button {
    width: parent.width
    height: 40
    text: "📁 Excel (CSV) Olarak Dışa Aktar"
    font.pixelSize: 13
    font.bold: true
    enabled: olcumId > 0

    onClicked: {
        database.csvDisaAktar(olcumId)
        console.log("CSV disa aktarildi.")
    }

    background: Rectangle {
        radius: 8
        color: parent.enabled ? (parent.hovered ? "#22c55e" : "#16a34a") : "#1e2a3f"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}
}

Rectangle {

```

```
width: parent.width
radius: 10
color: "#0a0e17"
border.color: "#1e2a3f"
border.width: 1
height: playbackIcerik.implicitHeight + 24
visible: olcumId > 0
```

```
Column {
    id: playbackIcerik
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.margins: 12
    spacing: 8

    Text {
        text: "OYNATMA (PLAYBACK)"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 10
        font.letterSpacing: 1
    }

    Text {
        text: strokeModeli.count > 0
            ? "Stroke " + (playbackAdimi + 1) + " / " + strokeModeli.count
            : "Oynatılacak stroke yok"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 12
    }

    Row {
        width: parent.width
        spacing: 8

        Button {
            width: 44
            height: 34
            text: oynatiliyor ? "||" : "▶"
            font.pixelSize: 14
            enabled: strokeModeli.count > 0

            onClicked: {
                if (oynatiliyor) {
                    oynatiliyor = false
                    playbackTimer.stop()
                } else {
                    if (playbackAdimi >= strokeModeli.count - 1) {
                        playbackAdimi = -1
                    }
                    oynatiliyor = true
                    playbackTimer.start()
                }
            }

            background: Rectangle {
                radius: 6
                color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") :
```

```

        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Repeater {
    model: [1, 2, 5, 10]

    Button {
        width: 44
        height: 34
        text: modelData + "x"
        font.pixelSize: 12

        onClicked: playbackHizi = modelData

        background: Rectangle {
            radius: 6
            color: playbackHizi === modelData ? "#3b82f6" : "#0f1420"
            border.color: "#1e2a3f"
            border.width: 1
        }

        contentItem: Text {
            text: parent.text
            color: playbackHizi === modelData ? "#ffffff" : "#9ca3af"
            font: parent.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }
}
}
}
}
}
}
}
}
}

Rectangle {
    id: panelAyraci
    width: 3
    height: parent.height
    color: "#05070b"

    Rectangle {
        anchors.right: parent.right
        anchors.top: parent.top
        anchors.bottom: parent.bottom
        width: 1
        color: "#403b82f6"
    }
}

```

```
}
```

```
Item {
```

```
    width: parent.width - ozetPaneli.width - panelAyraci.width
```

```
    height: parent.height
```

```
    Rectangle {
```

```
        id: grafikKutusu
```

```
        anchors.top: parent.top
```

```
        anchors.left: parent.left
```

```
        anchors.right: parent.right
```

```
        anchors.margins: 16
```

```
        height: parent.height * 0.55
```

```
        radius: 10
```

```
        color: "#0f1420"
```

```
        border.color: "#1e2a3f"
```

```
        border.width: 1
```

```
        Item {
```

```
            id: grafikBaslik
```

```
            anchors.top: parent.top
```

```
            anchors.left: parent.left
```

```
            anchors.right: parent.right
```

```
            anchors.margins: 14
```

```
            height: 20
```

```
            Text {
```

```
                anchors.left: parent.left
```

```
                anchors.verticalCenter: parent.verticalCenter
```

```
                text: "P-Q Dağılım Grafiği"
```

```
                color: "#9ca3af"
```

```
                font.family: "Segoe UI"
```

```
                font.pixelSize: 13
```

```
                font.bold: true
```

```
            }
```

```
        Row {
```

```
            anchors.right: parent.right
```

```
            anchors.verticalCenter: parent.verticalCenter
```

```
            spacing: 16
```

```
        Row {
```

```
            spacing: 6
```

```
            anchors.verticalCenter: parent.verticalCenter
```

```
            Rectangle {
```

```
                width: 8
```

```
                height: 8
```

```
                radius: 4
```

```
                color: "#3b82f6"
```

```
                anchors.verticalCenter: parent.verticalCenter
```

```
            }
```

```
            Text {
```

```
                text: "Ölçüm Noktaları"
```

```
                color: "#6b7280"
```

```
                font.family: "Segoe UI"
```

```
                font.pixelSize: 11
```

```
                anchors.verticalCenter: parent.verticalCenter
```

```
            }
```

```

    }

    Row {
        spacing: 6
        anchors.verticalCenter: parent.verticalCenter

        Rectangle {
            width: 12
            height: 2
            color: "#e8a020"
            anchors.verticalCenter: parent.verticalCenter
        }

        Text {
            text: "Regresyon Doğrusu"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 11
            anchors.verticalCenter: parent.verticalCenter
        }
    }

    Row {
        spacing: 6
        anchors.verticalCenter: parent.verticalCenter

        Rectangle {
            width: 8
            height: 8
            radius: 4
            color: "#e8a020"
            anchors.verticalCenter: parent.verticalCenter
        }

        Text {
            text: "Oynatma"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 11
            anchors.verticalCenter: parent.verticalCenter
        }
    }
}

}

ChartView {
    anchors.top: grafikBaslik.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 4
    anchors.margins: 8
    backgroundColor: "transparent"
    legend.visible: false
    antialiasing: true

    ValueAxis {
        id: qEkseni
        min: 0
        max: 20
    }
}

```

```

        titleText: "Debi (Q) m³/h"
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    ValueAxis {
        id: pEkseni
        min: 0
        max: 100
        titleText: "Basınç (P) mbar"
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    ScatterSeries {
        id: pqSerisi
        axisX: qEkseni
        axisY: pEkseni
        color: "#3b82f6"
        markerSize: 10
    }

    LineSeries {
        id: regresyonCizgisi
        axisX: qEkseni
        axisY: pEkseni
        color: "#e8a020"
        width: 2
    }

    ScatterSeries {
        id: oynatmaSerisi
        axisX: qEkseni
        axisY: pEkseni
        color: "#e8a020"
        markerSize: 16
    }
}

Rectangle {
    anchors.top: grafikKutusu.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.margins: 16
    anchors.topMargin: 8
    radius: 10
    color: "#0f1420"
    border.color: "#1e2a3f"
    border.width: 1

    Text {
        anchors.top: parent.top

```

```

        anchors.left: parent.left
        anchors.margins: 14
        text: "Stroke Tablosu"
        color: "#9ca3af"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }

    Row {
        id: tabloBasligi
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.topMargin: 36
        anchors.leftMargin: 12
        anchors.rightMargin: 12

        Text { width: parent.width * 0.25; text: "STROKE"; color: "#6b7280"; font.pixelSize: 11;
font.bold: true; font.letterSpacing: 1 }
        Text { width: parent.width * 0.25; text: "P (mbar)"; color: "#6b7280"; font.pixelSize:
11; font.bold: true; font.letterSpacing: 1 }
        Text { width: parent.width * 0.25; text: "Q (m³/h)"; color: "#6b7280"; font.pixelSize:
11; font.bold: true; font.letterSpacing: 1 }
        Text { width: parent.width * 0.25; text: "DURUM"; color: "#6b7280"; font.pixelSize: 11;
font.bold: true; font.letterSpacing: 1 }
    }

    Rectangle {
        anchors.top: tabloBasligi.bottom
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.topMargin: 8
        anchors.leftMargin: 12
        anchors.rightMargin: 12
        height: 1
        color: "#1e2a3f"
    }

    ListView {
        anchors.top: tabloBasligi.bottom
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.margins: 12
        anchors.topMargin: 14
        clip: true

        model: ListModel {
            id: strokeModeli
        }

        delegate: Rectangle {
            width: parent.width
            height: 32
            radius: 6
            color: index == playbackAdimi ? "#1e2a3f" : (index % 2 == 0 ? "transparent" :
"#0a0e17")

            Row {

```

```

        Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter;
text: stroke; color: "#dce8f5"; font.pixelSize: 12 }
            Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter;
text: p.toFixed(1); color: "#dce8f5"; font.pixelSize: 12 }
                Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter;
text: q.toFixed(2); color: "#dce8f5"; font.pixelSize: 12 }

        Item {
            width: parent.width * 0.25
            height: parent.height

            Rectangle {
                anchors.verticalCenter: parent.verticalCenter
                width: durumMetni.implicitWidth + 16
                height: 20
                radius: 10
                color: gecerli ? "#123321" : "#2a1414"

                Text {
                    id: durumMetni
                    anchors.centerIn: parent
                    text: gecerli ? "Geçerli" : "Hatalı"
                    color: gecerli ? "#4ade80" : "#f87171"
                    font.pixelSize: 11
                    font.bold: true
                }
            }
        }
    }
}

```